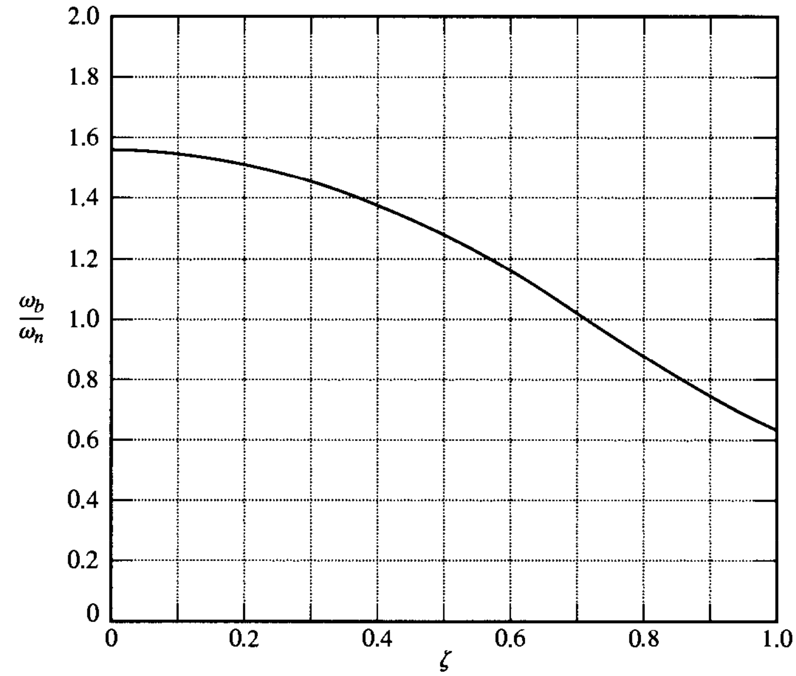



**Figure 8–125**
Curve $\omega_b/\omega_n$ versus $\zeta$, where $\omega_b$ is the bandwidth.

Consider the frequency response of this system. Plot a polar locus of $G(j\omega)/K$. Then determine the value of gain $K$ such that the resonant peak magnitude $M_r$, of the closed-loop frequency response is 2.

**Solution.** A plot of $G(j\omega)/K$ is shown in Figure 8–126. The value of angle $\psi$ corresponding to $M_r = 2$ obtained from Equation (8–18) is

$$\psi = \sin^{-1}\frac{1}{2} = 30°$$

Hence, we draw the line $\overline{OP}$ that passes through the origin and makes an angle of 30° with the negative real axis as shown in Figure 8–126. We then draw the circle that is tangent to both the $G(j\omega)/K$ locus and line $\overline{OP}$. Define the point where the circle and line $\overline{OP}$ are tangent as point $P$. The perpendicular line drawn from point $P$ intersects the negative real axis at $(-0.445,0)$. Then the gain $K$ is determined as

$$K = \frac{1}{0.445} = 2.247$$

From Figure 8–126 we notice that the resonant frequency is approximately $\omega = 0.83$ rad/sec.

**A-8-23.** Figure 8–127 shows a block diagram of a chemical reactor system. Draw a Bode diagram of $G(j\omega)$. Also, draw the $G(j\omega)$ locus on the Nichols chart. From the Nichols diagram, read magnitudes and phase angles of the closed loop frequency response and then plot the Bode diagram of the closed-loop system, $G(j\omega)/[1 + G(j\omega)]$.

**Solution.** Noting that

$$G(s) = \frac{80e^{-0.1s}}{s(s+4)(s+10)} = \frac{2e^{-0.1s}}{s(0.25s+1)(0.1s+1)}$$

we have

$$G(j\omega) = \frac{2e^{-0.1j\omega}}{j\omega(0.25j\omega+1)(0.1j\omega+1)}$$

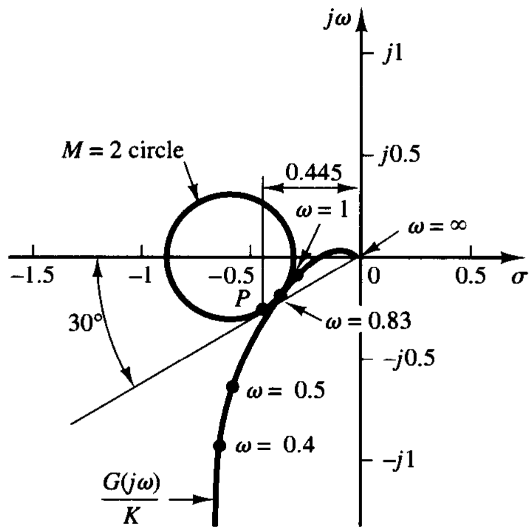



**Figure 8–126**
Plot of $G(j\omega)/K$
of the system
considered in Problem A-8-22.

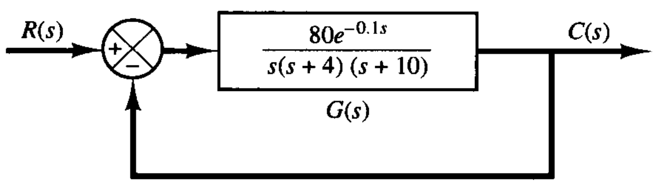



**Figure 8–127**
Block diagram of a chemical
reactor system.

The phase angle of the transport lag $e^{-0.1j\omega}$ is

$$\underline{/e^{-0.1j\omega}} = \underline{/\cos(0.1\omega) - j\sin(0.1\omega)} = -0.1\omega \quad \text{(rad)}$$
$$= -5.73\omega \quad \text{(degrees)}$$

The Bode diagram of $G(j\omega)$ is shown in Figure 8–128.

Next, by reading magnitudes and phase angles of $G(j\omega)$ for various values of $\omega$, it is possible to plot the gain versus phase plot on a Nichols chart. Figure 8–129 shows such a $G(j\omega)$ locus superimposed on the Nichols chart. From this diagram, magnitudes and phase angles of the closed-loop system at various frequency points can be read. Figure 8–130 depicts the Bode diagram of the closed-loop frequency response $G(j\omega)/[1 + G(j\omega)]$.

**A-8-24.** A Bode diagram of the open-loop transfer function $G(s)$ of a unity-feedback control system is shown in Figure 8–131. It is known that the open-loop transfer function is minimum phase. From the diagram it can be seen that there is a pair of complex-conjugate poles at $\omega = 2$ rad/sec. Determine the damping ratio of the quadratic term involving these complex-conjugate poles. Also, determine the transfer function $G(s)$.

**Solution.** Referring to Figure 8–8 and examining the Bode diagram of Figure 8–131, we find the damping ratio $\zeta$ and undamped natural frequency $\omega_n$ of the quadratic term to be

$$\zeta = 0.1, \qquad \omega_n = 2 \text{ rad/sec}$$

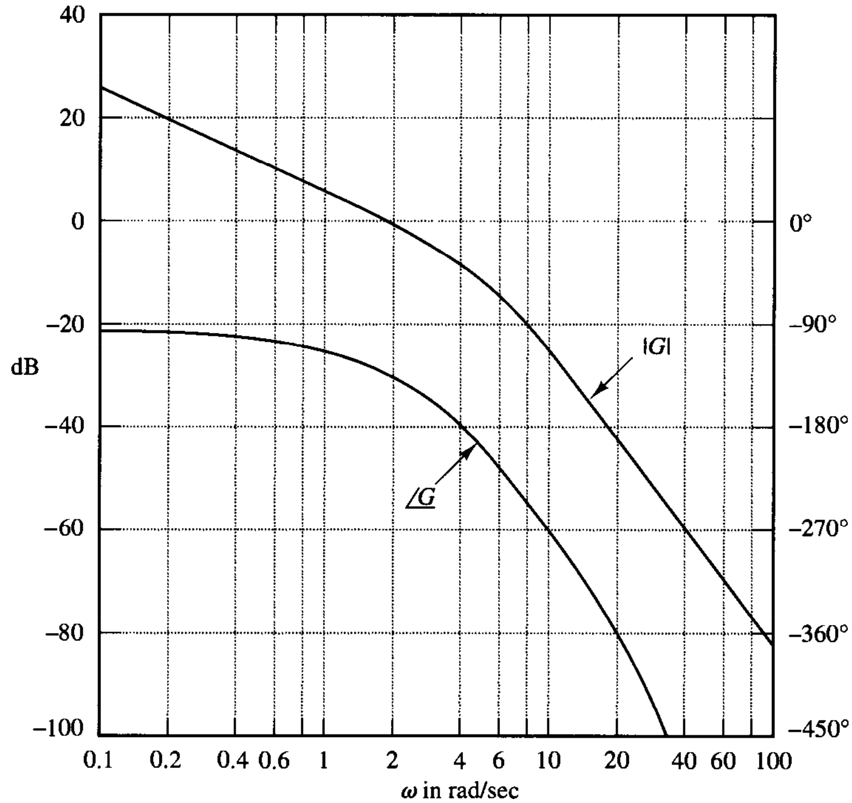



**Figure 8–128** Bode diagram of $G(j\omega)$ of the system shown in Figure 8–127.

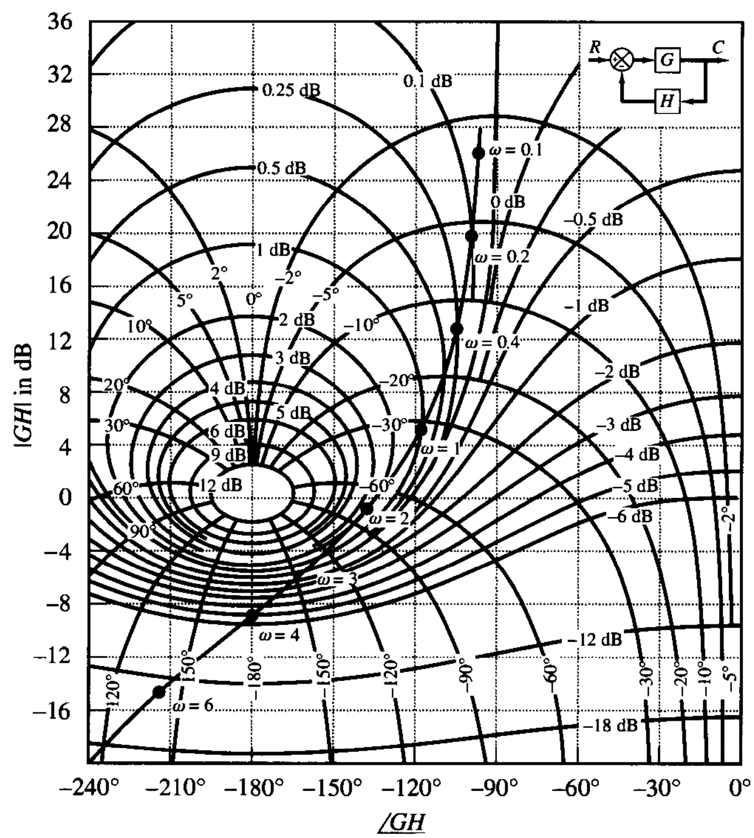



**Figure 8–129** $G(j\omega)$ locus superimposed on Nichols chart (Problem A-8-23).

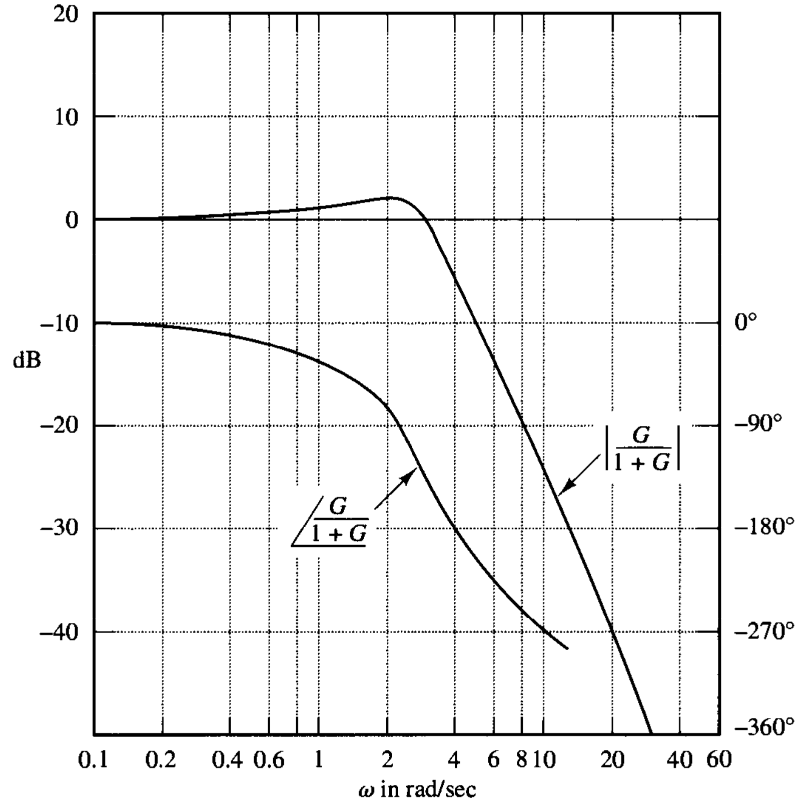



**Figure 8–130**
Bode diagram of the closed-loop frequency response (Problem A-8-23).

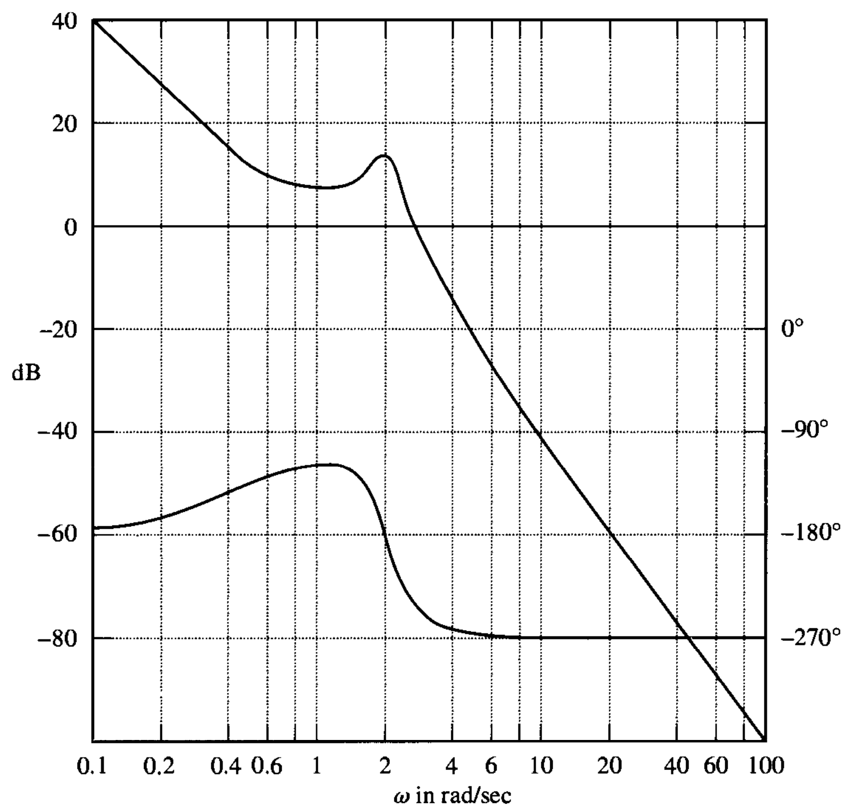



**Figure 8–131**
Bode diagram of the open-loop transfer function of a unity-feedback control system.

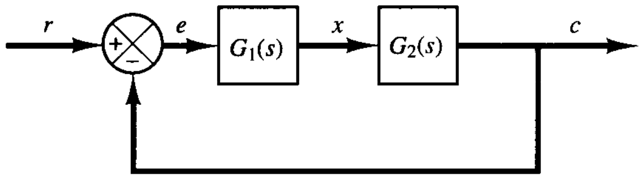


**Figure 8–132**
Control system.

Noting that there is another corner frequency at $\omega = 0.5$ rad/sec and the slope of the magnitude curve in the low-frequency region is $-40$ dB/decade, $G(j\omega)$ can be tentatively determined as follows:

$$G(j\omega) = \frac{K\left(\dfrac{j\omega}{0.5} + 1\right)}{(j\omega)^2\left[\left(\dfrac{j\omega}{2}\right)^2 + 0.1(j\omega) + 1\right]}$$

Since from Figure 8–131 we find $|G(j0.1)| = 40$ dB, the gain value $K$ can be determined as unity. Also, the calculated phase curve, $\angle G(j\omega)$ versus $\omega$, agrees with the given phase curve. Hence, the transfer function $G(s)$ can be determined as

$$G(s) = \frac{4(2s + 1)}{s^2(s^2 + 0.4s + 4)}$$

**A-8-25.** A closed-loop control system may include an unstable element within the loop. When the Nyquist stability criterion is to be applied to such a system, the frequency-response curves for the unstable element must be obtained.

How can we obtain experimentally the frequency-response curves for such an unstable element? Suggest a possible approach to the experimental determination of the frequency response of an unstable linear element.

**Solution.** One possible approach is to measure the frequency-response characteristics of the unstable element by using it as a part of a stable system.

Consider the system shown in Figure 8–132. Suppose that the element $G_1(s)$ is unstable. The complete system may be made stable by choosing a suitable linear element $G_2(s)$. We apply a sinusoidal signal at the input. At steady state, all signals in the loop will be sinusoidal. We measure the signals $e(t)$, the input to the unstable element, and $x(t)$, the output of the unstable element. By changing the frequency [and possibly the amplitude for the convenience of measuring $e(t)$ and $x(t)$] of the input sinusoid and repeating this process, it is possible to obtain the frequency response of the unstable linear element.

## PROBLEMS

**B-8-1.** Consider the unity-feedback system with the open-loop transfer functions.

$$G(s) = \frac{10}{s + 1}$$

Obtain the steady-state output of the system when it is subjected to each of the following inputs:

(a) $r(t) = \sin(t + 30°)$
(b) $r(t) = 2\cos(2t - 45°)$
(c) $r(t) = \sin(t + 30°) - 2\cos(2t - 45°)$

**B-8-2.** Consider the system whose closed-loop transfer function is

$$\frac{C(s)}{R(s)} = \frac{K(T_2 s + 1)}{T_1 s + 1}$$

Obtain the steady-state output of the system when it is subjected to the input $r(t) = R\sin\omega t$.

**B-8-3.** Sketch the Bode diagrams of the following three transfer functions:

(a) $G(s) = \dfrac{T_1 s + 1}{T_2 s + 1} \qquad (T_1 > T_2 > 0)$

(b) $G(s) = \dfrac{T_1 s - 1}{T_2 s + 1} \qquad (T_1 > T_2 > 0)$

(c) $G(s) = \dfrac{-T_1 s + 1}{T_2 s + 1} \qquad (T_1 > T_2 > 0)$

**B-8-4.** Plot the Bode diagram of

$$G(s) = \frac{10(s^2 + 0.4s + 1)}{s(s^2 + 0.8s + 9)}$$

**B-8-5.** Given

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

show that

$$|G(j\omega_n)| = \frac{1}{2\zeta}$$

**B-8-6.** Consider a unity-feedback control system with the following open-loop transfer function:

$$G(s) = \frac{s + 0.5}{s^3 + s^2 + 1}$$

This is a nonminimum-phase system. Two of the three open-loop poles are located in the right-half $s$ plane as follows:

$$\text{Open-loop poles at } s = -1.4656$$
$$s = 0.2328 + j0.7926$$
$$s = 0.2328 - j0.7926$$

Plot the Bode diagram of $G(s)$ with MATLAB. Explain why the phase-angle curve starts from 0° and approaches +180°.

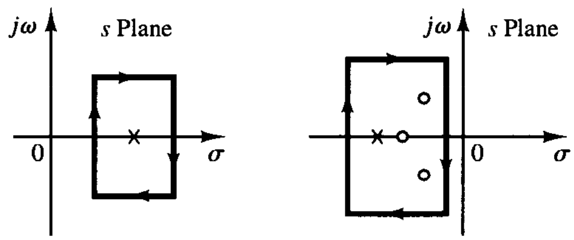
**B-8-7.** Sketch the polar plots of the open-loop transfer function

$$G(s)H(s) = \frac{K(T_a s + 1)(T_b s + 1)}{s^2(Ts + 1)}$$

for the following two cases:

(a) $T_a > T > 0, \qquad T_b > T > 0$
(b) $T > T_a > 0, \qquad T > T_b > 0$

**B-8-8.** The pole–zero configurations of complex functions $F_1(s)$ and $F_2(s)$ are shown in Figures 8–133 (a) and (b), respectively. Assume that the closed contours in the $s$ plane are those shown in Figures 8–133 (a) and (b). Sketch quali-




(a) (b)

**Figure 8–133** (a) $s$-Plane representation of complex function $F_1(s)$ and a closed contour; (b) $s$-Plane representation of complex function $F_2(s)$ and a closed contour.

tatively the corresponding closed contours in the $F_1(s)$ plane and $F_2(s)$ plane.

**B-8-9.** Draw a Nyquist locus for the unity feedback control system with the open loop transfer function

$$G(s) = \frac{K(1 - s)}{s + 1}$$

Using the Nyquist stability criterion, determine the stability of the closed loop system.

**B-8-10.** A system with the open-loop transfer function

$$G(s)H(s) = \frac{K}{s^2(T_1 s + 1)}$$

is inherently unstable. This system can be stabilized by adding derivative control. Sketch the polar plots for the open-loop transfer function with and without derivative control.

**B-8-11.** Consider the closed-loop system with the following open-loop transfer function:

$$G(s)H(s) = \frac{10K(s + 0.5)}{s^2(s + 2)(s + 10)}$$

Plot both the direct and inverse polar plots of $G(s)H(s)$ with $K = 1$ and $K = 10$. Apply the Nyquist stability criterion to the plots and determine the stability of the system with these values of $K$.

**B-8-12.** Consider the closed-loop system whose open-loop transfer function is

$$G(s)H(s) = \frac{Ke^{-2s}}{s}$$

Find the maximum value of $K$ for which the system is stable.

**B-8-13.** Draw a Nyquist plot of the following $G(s)$:

$$G(s) = \frac{1}{s(s^2 + 0.8s + 1)}$$

**B-8-14.** Consider a unity-feedback control system with the following open-loop transfer function:

$$G(s) = \frac{1}{s^3 + 0.2s^2 + s + 1}$$

Draw a Nyquist plot of $G(s)$ and examine the stability of the system.

**B-8-15.** Consider a unity-feedback control system with the following open-loop transfer function:

$$G(s) = \frac{s^2 + 2s + 1}{s^3 + 0.2s^2 + s + 1}$$

Draw a Nyquist plot of $G(s)$ and examine the stability of the closed-loop system.

**B-8-16.** Consider the system defined by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 6.5 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

There are four individual Nyquist plots involved in this system. Draw two Nyquist plots for the input $u_1$ in one diagram and two Nyquist plots for the input $u_2$ in another diagram. Write a MATLAB program to obtain these two diagrams.

**B-8-17.** Referring to Problem B-8-16, it is desired to plot only $Y_1(j\omega)/U_1(j\omega)$ for $\omega > 0$. Write a MATLAB program to produce such a plot.

If it is desired to plot $Y_1(j\omega)/U_1(j\omega)$ for $-\infty < \omega < \infty$, what changes must be made in the MATLAB program?

**B-8-18.** Consider the unity-feedback control system whose open-loop transfer function is

$$G(s) = \frac{as + 1}{s^2}$$

Determine the value of $a$ so that the phase margin is 45°.

**B-8-19.** Consider the system shown in Figure 8–134. Draw a Bode diagram of the open-loop transfer function $G(s)$. Determine the phase margin and gain margin.

**B-8-20.** Consider the system shown in Figure 8–135. Draw a Bode diagram of the open-loop transfer function $G(s)$. Determine the phase margin and gain margin.

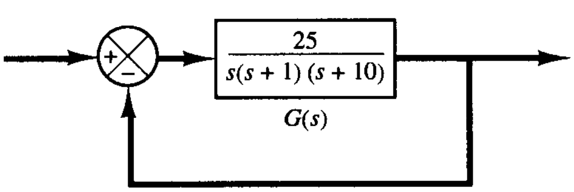



**Figure 8–134** Control system.

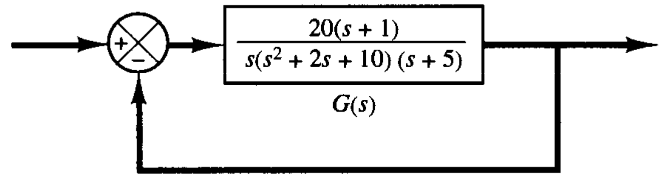



**Figure 8–135** Control system.

**B-8-21.** Consider a unity-feedback control system with the open-loop transfer function

$$G(s) = \frac{K}{s(s^2 + s + 4)}$$

Determine the value of the gain $K$ such that the phase margin is 50°. What is the gain margin with this gain $K$?

**B-8-22.** Consider the system shown in Figure 8–136. Draw a Bode diagram of the open-loop transfer function and determine the value of the gain $K$ such that the phase margin is 50°. What is the gain margin of this system with this gain $K$?

**B-8-23.** Consider a unity-feedback control system whose open-loop transfer function is

$$G(s) = \frac{K}{s(s^2 + s + 0.5)}$$

Determine the value of the gain $K$ such that the resonant peak magnitude in the frequency response is 2 dB, or $M_r = 2$ dB.

**B-8-24.** Figure 8–137 shows a block diagram of a process control system. Determine the range of gain $K$ for stability.

**B-8-25.** Consider a closed-loop system whose open-loop transfer function is

$$G(s)H(s) = \frac{Ke^{-Ts}}{s(s + 1)}$$

Determine the maximum value of the gain $K$ for stability as a function of dead time $T$.

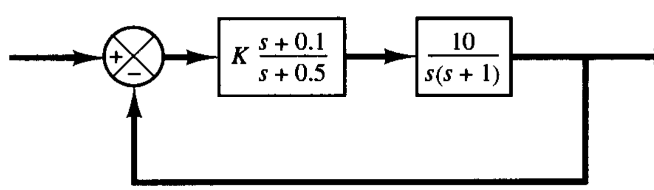



**Figure 8–136** Control system.

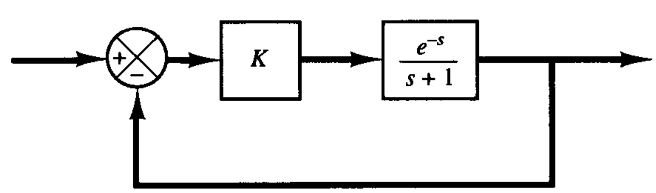



**Figure 8–137** Process control system.

**B-8-26.** Sketch the polar plot of

$$G(s) = \frac{(Ts)^2 - 6(Ts) + 12}{(Ts)^2 + 6(Ts) + 12}$$

Show that, for the frequency range $0 < \omega T < 2\sqrt{3}$, this equation gives a good approximation to the transfer function of transport lag, $e^{-Ts}$.

**B-8-27.** Figure 8–138 shows a Bode diagram of a transfer function $G(s)$. Determine this transfer function.

**B-8-28.** The experimentally determined Bode diagram of a system $G(j\omega)$ is shown in Figure 8–139. Determine the transfer function $G(s)$.

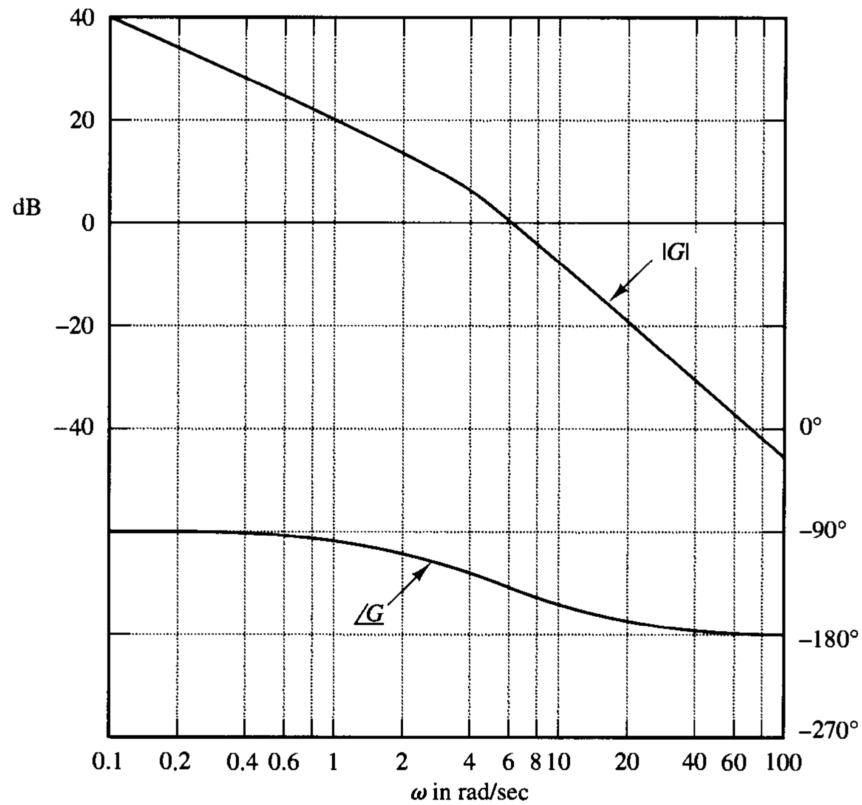



**Figure 8–138** Bode diagram of a transfer function $G(s)$.

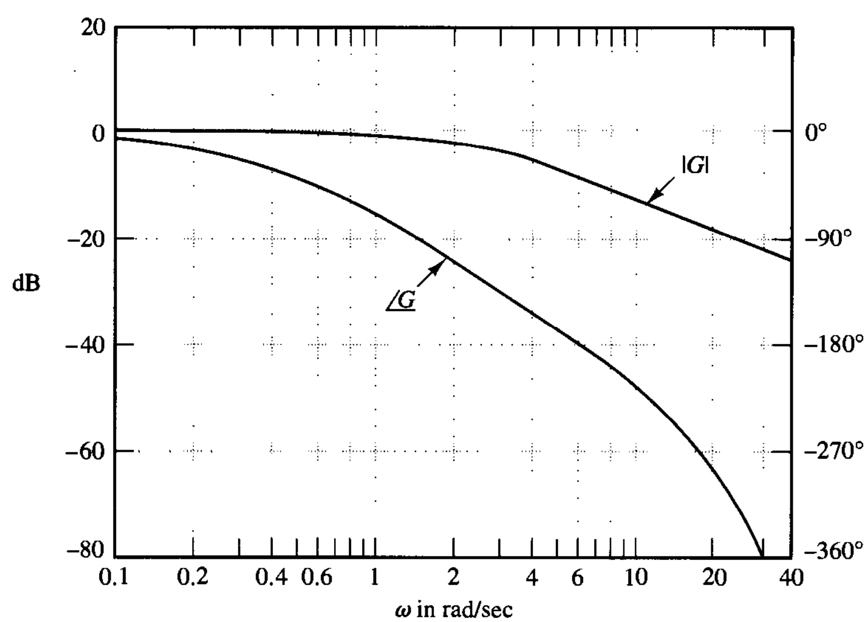



**Figure 8–139** Experimentally determined Bode diagram of a system.

# 9

# Control Systems Design by Frequency Response

## 9–1 INTRODUCTION

The primary objective of this chapter is to present procedures for the design and compensation of single-input–single-output, linear, time-invariant control systems by the frequency-response approach.

**Frequency-response approach to control system design.** It is important to note that in a control system design, transient-response performance is usually most important. In the frequency-response approach, we specify the transient-response performance in an indirect manner. That is, the transient-response performance is specified in terms of the phase margin, gain margin, resonant peak magnitude (they give a rough estimate of the system damping); the gain crossover frequency, resonant frequency, bandwidth (they give a rough estimate of the speed of transient response); and static error constants (they give the steady-state accuracy). Although the correlation between the transient response and frequency response is indirect, the frequency domain specifications can be conveniently met in the Bode diagram approach.

After the open loop has been designed by the frequency-response method, the closed-loop poles and zeros can be determined. The transient-response characteristics must be checked to see whether the designed system satisfies the requirements in the time domain. If it does not, then the compensator must be modified and the analysis repeated until a satisfactory result is obtained.

Design in the frequency domain is simple and straightforward. The frequency-response plot indicates clearly the manner in which the system should be modified, although the exact quantitative prediction of the transient-response characteristics cannot be made. The frequency-response approach can be applied to systems or components whose dynamic characteristics are given in the form of frequency-response data. Note that because of difficulty in deriving the equations governing certain components, such as pneumatic and hydraulic components, the dynamic characteristics of such components are usually determined experimentally through frequency-response tests. The experimentally obtained frequency-response plots can be combined easily with other such plots when the Bode diagram approach is used. Note also that in dealing with high-frequency noises we find that the frequency-response approach is more convenient than other approaches.

There are basically two approaches in the frequency-domain design. One is the polar plot approach and the other is the Bode diagram approach. When a compensator is added, the polar plot does not retain the original shape, and, therefore, we need to draw a new polar plot, which will take time and is thus inconvenient. On the other hand, a Bode diagram of the compensator can be simply added to the original Bode diagram, and thus plotting the complete Bode diagram is a simple matter. Also, if the open-loop gain is varied, the magnitude curve is shifted up or down without changing the slope of the curve, and the phase curve remains the same. For design purposes, therefore, it is best to work with the Bode diagram.

A common approach to the Bode diagram is that we first adjust the open-loop gain so that the requirement on the steady-state accuracy is met. Then the magnitude and phase curves of the uncompensated open loop (with the open-loop gain just adjusted) is plotted. If the specifications on the phase margin and gain margin are not satisfied, then a suitable compensator that will reshape the open-loop transfer function is determined. Finally, if there are any other requirements to be met, we try to satisfy them, unless some of them are contradictory to each other.

**Information obtainable from open-loop frequency response.** The low-frequency region (the region far below the gain crossover frequency) of the locus indicates the steady-state behavior of the closed-loop system. The medium-frequency region (the region near the $-1 + j0$ point) of the locus indicates relative stability. The high-frequency region (the region far above the gain crossover frequency) indicates the complexity of the system.

**Requirements on open-loop frequency response.** We might say that, in many practical cases, compensation is essentially a compromise between steady-state accuracy and relative stability.

To have a high value of the velocity error constant and yet satisfactory relative stability, we find it necessary to reshape the open-loop frequency-response curve.

The gain in the low-frequency region should be large enough, and also, near the gain crossover frequency, the slope of the log-magnitude curve in the Bode diagram should be $-20$ dB/decade. This slope should extend over a sufficiently wide frequency band to assure a proper phase margin. For the high-frequency region, the gain should be attenuated as rapidly as possible to minimize the effects of noise.

Examples of generally desirable and undesirable open-loop and closed-loop frequency-response curves are shown in Figure 9–1.

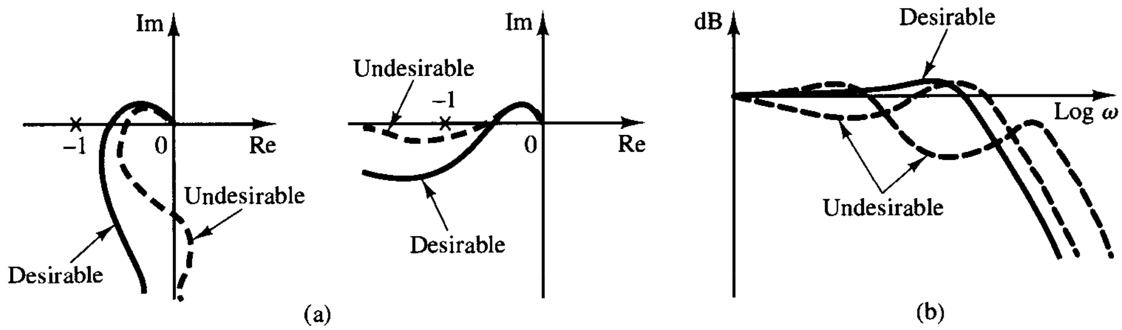



**Figure 9–1**
(a) Examples of desirable and undesirable open-loop frequency-response curves; (b) examples of desirable and undesirable closed-loop frequency-response curves.

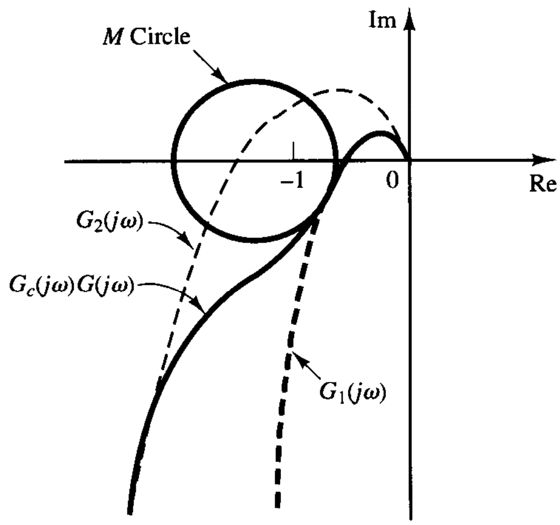



**Figure 9–2**
Reshaping of the open-loop frequency-response curve.

Referring to Figure 9–2, we see that the reshaping of the open-loop frequency-response curve may be done if the high-frequency portion of the locus follows the $G_1(j\omega)$ locus, while the low-frequency portion of the locus follows the $G_2(j\omega)$ locus. The reshaped locus $G_c(j\omega)G(j\omega)$ should have reasonable phase and gain margins or should be tangent to a proper $M$ circle, as shown.

**Basic characteristics of lead, lag, and lag–lead compensation.** Lead compensation essentially yields an appreciable improvement in transient response and a small change in steady-state accuracy. It may accentuate high-frequency noise effects. Lag compensation, on the other hand, yields an appreciable improvement in steady-state accuracy at the expense of increasing the transient-response time. Lag compensation will suppress the effects of high-frequency noise signals. Lag–lead compensation combines the characteristics of both lead compensation and lag compensation. The use of a lead or lag compensator raises the order of the system by 1 (unless cancellation occurs between the zero of the compensator and a pole of the uncompensated open-loop transfer function). The use of a lag–lead compensator raises the order of the system by 2 [unless cancellation occurs between zero(s) of the lag–lead compensator and pole(s) of the uncompensated open-loop transfer function], which means that the system becomes more complex and it is more difficult to control the transient response behavior. The particular situation determines the type of compensation to be used.

**Outline of the chapter.** Section 9–1 has presented introductory material. Section 9–2 discusses lead compensation by the Bode diagram approach and, Section 9–3 treats lag compensation by the Bode diagram approach. Section 9–4 discusses lag–lead compensation techniques based on the Bode diagram approach. Section 9–5 gives concluding comments on the frequency-response approach to the control systems design.

## 9–2 LEAD COMPENSATION

We shall first examine the frequency characteristics of the lead compensator. Then we present a design technique for the lead compensator by use of the Bode diagram.

**Characteristics of lead compensators.** Consider a lead compensator having the following transfer function:

$$K_c\alpha\frac{Ts+1}{\alpha Ts+1} = K_c\frac{s+\dfrac{1}{T}}{s+\dfrac{1}{\alpha T}} \qquad (0<\alpha<1)$$

It has a zero at $s=-1/T$ and a pole at $s=-1/(\alpha T)$. Since $0<\alpha<1$, we see that the zero is always located to the right of the pole in the complex plane. Note that for a small value of $\alpha$ the pole is located far to the left. The minimum value of $\alpha$ is limited by the physical construction of the lead compensator. The minimum value of $\alpha$ is usually taken to be about 0.05. (This means that the maximum phase lead that may be produced by a lead compensator is about 65°.)

Figure 9–3 shows the polar plot of

$$K_c\alpha\frac{j\omega T+1}{j\omega\alpha T+1} \qquad (0<\alpha<1)$$

with $K_c = 1$. For a given value of $\alpha$, the angle between the positive real axis and the tangent line drawn from the origin to the semicircle gives the maximum phase lead angle,

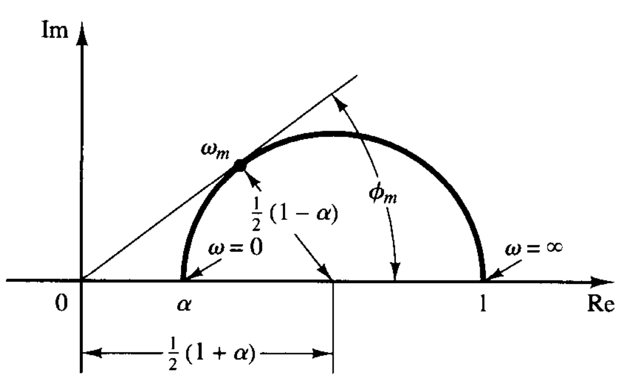



**Figure 9–3**
Polar plot of a lead compensator $\alpha(j\omega T+1)/(j\omega\alpha T+1)$, where $0<\alpha<1$.

$\phi_m$. We shall call the frequency at the tangent point $\omega_m$. From Figure 9–3 the phase angle at $\omega = \omega_m$ is $\phi_m$, where

$$\sin \phi_m = \frac{\dfrac{1-\alpha}{2}}{\dfrac{1+\alpha}{2}} = \frac{1-\alpha}{1+\alpha} \tag{9–1}$$

Equation (9–1) relates the maximum phase lead angle and the value of $\alpha$.

Figure 9–4 shows the Bode diagram of a lead compensator when $K_c = 1$ and $\alpha = 0.1$. The corner frequencies for the lead compensator are $\omega = 1/T$ and $\omega = 1/(\alpha T) = 10/T$. By examining Figure 9–4, we see that $\omega_m$ is the geometric mean of the two corner frequencies, or

$$\log \omega_m = \frac{1}{2}\left(\log \frac{1}{T} + \log \frac{1}{\alpha T}\right)$$

Hence,

$$\omega_m = \frac{1}{\sqrt{\alpha}\,T} \tag{9–2}$$

As seen from Figure 9–4, the lead compensator is basically a high-pass filter. (The high frequencies are passed, but low frequencies are attenuated.)

**Lead compensation techniques based on the frequency-response approach.** The primary function of the lead compensator is to reshape the frequency-response curve to provide sufficient phase-lead angle to offset the excessive phase lag associated with the components of the fixed system.

Consider the system shown in Figure 9–5. Assume that the performance specifications are given in terms of phase margin, gain margin, static velocity error constants, and so on. The procedure for designing a lead compensator by the frequency-response approach may be stated as follows:

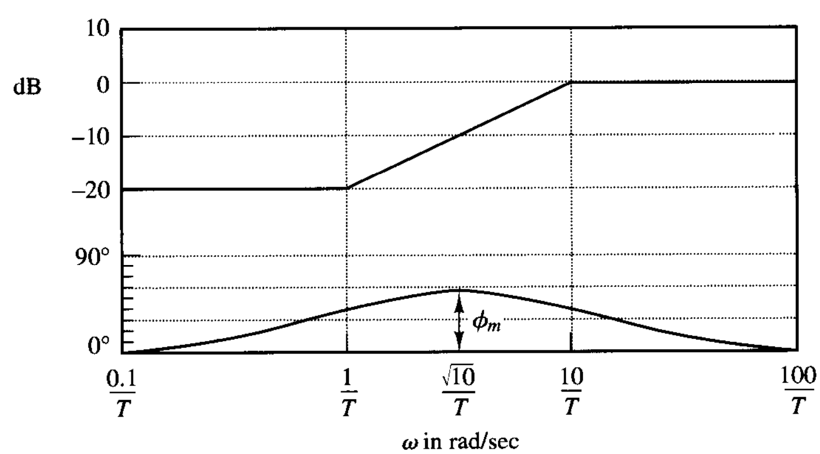



**Figure 9–4** Bode diagram of a lead compensator $\alpha(j\omega T+1)/(j\omega\alpha T+1)$, where $\alpha = 0.1$.

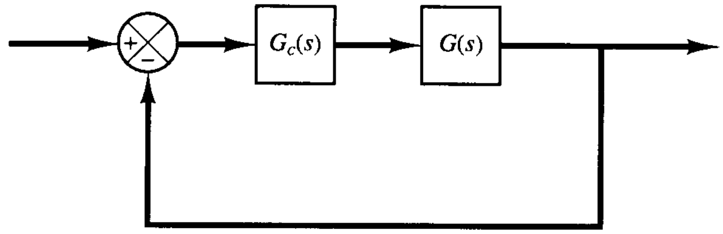



**Figure 9–5**
Control system.

**1.** Assume the following lead compensator:

$$G_c(s) = K_c\alpha\frac{Ts+1}{\alpha Ts+1} = K_c\frac{s+\frac{1}{T}}{s+\frac{1}{\alpha T}} \qquad (0 < \alpha < 1)$$

Define

$$K_c\alpha = K$$

Then

$$G_c(s) = K\frac{Ts+1}{\alpha Ts+1}$$

The open-loop transfer function of the compensated system is

$$G_c(s)G(s) = K\frac{Ts+1}{\alpha Ts+1}G(s) = \frac{Ts+1}{\alpha Ts+1}KG(s) = \frac{Ts+1}{\alpha Ts+1}G_1(s)$$

where

$$G_1(s) = KG(s)$$

Determine gain $K$ to satisfy the requirement on the given static error constant.

**2.** Using the gain $K$ thus determined, draw a Bode diagram of $G_1(j\omega)$, the gain-adjusted but uncompensated system. Evaluate the phase margin.

**3.** Determine the necessary phase lead angle $\phi$ to be added to the system.

**4.** Determine the attenuation factor $\alpha$ by use of Equation (9–1). Determine the frequency where the magnitude of the uncompensated system $G_1(j\omega)$ is equal to $-20\log(1/\sqrt{\alpha})$. Select this frequency as the new gain crossover frequency. This frequency corresponds to $\omega_m = 1/(\sqrt{\alpha}T)$, and the maximum phase shift $\phi_m$ occurs at this frequency.

**5.** Determine the corner frequencies of the lead compensator as follows:

$$\text{Zero of lead compensator:} \quad \omega = \frac{1}{T}$$

$$\text{Pole of lead compensator:} \quad \omega = \frac{1}{\alpha T}$$

**6.** Using the value of $K$ determined in step 1 and that of $\alpha$ determined in step 4, calculate constant $K_c$ from

$$K_c = \frac{K}{\alpha}$$

7. Check the gain margin to be sure it is satisfactory. If not, repeat the design process by modifying the pole–zero location of the compensator until a satisfactory result is obtained.

**EXAMPLE 9–1** Consider the system shown in Figure 9–6. The open-loop transfer function is

$$G(s) = \frac{4}{s(s+2)}$$

It is desired to design a compensator for the system so that the static velocity error constant $K_v$ is 20 sec$^{-1}$, the phase margin is at least 50°, and the gain margin is at least 10 dB.

We shall use a lead compensator of the form

$$G_c(s) = K_c\alpha \frac{Ts+1}{\alpha Ts+1} = K_c \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}}$$

The compensated system will have the open-loop transfer function $G_c(s)G(s)$.

Define

$$G_1(s) = KG(s) = \frac{4K}{s(s+2)}$$

where $K = K_c\alpha$.

The first step in the design is to adjust the gain $K$ to meet the steady-state performance specification or to provide the required static velocity error constant. Since this constant is given as 20 sec$^{-1}$, we obtain

$$K_v = \lim_{s\to 0} sG_c(s)G(s) = \lim_{s\to 0} s\frac{Ts+1}{\alpha Ts+1}G_1(s) = \lim_{s\to 0}\frac{s4K}{s(s+2)} = 2K = 20$$

or

$$K = 10$$

With $K = 10$, the compensated system will satisfy the steady-state requirement.

We shall next plot the Bode diagram of

$$G_1(j\omega) = \frac{40}{j\omega(j\omega+2)} = \frac{20}{j\omega(0.5j\omega+1)}$$

Figure 9–7 shows the magnitude and phase angle curves of $G_1(j\omega)$. From this plot, the phase and gain margins of the system are found to be 17° and $+\infty$ dB, respectively. (A phase margin of 17° implies that the system is quite oscillatory. Thus, satisfying the specification on the steady state

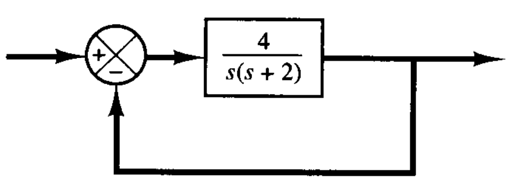



**Figure 9–6**
Control system.

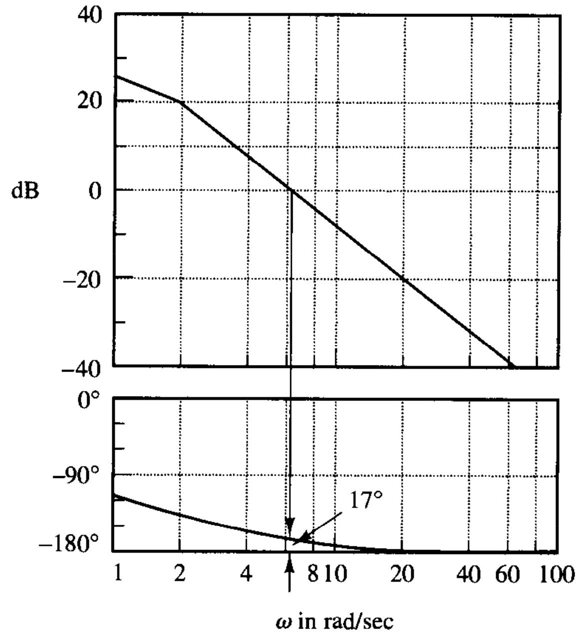



**Figure 9–7**
Bode diagram for
$G_1(j\omega) = 10G(j\omega) = 40/[j\omega(j\omega + 2)]$

yields a poor transient-response performance.) The specification calls for a phase margin of at least 50°. We thus find the additional phase lead necessary to satisfy the relative stability requirement is 33°. To achieve a phase margin of 50° without decreasing the value of $K$, the lead compensator must contribute the required phase angle.

Noting that the addition of a lead compensator modifies the magnitude curve in the Bode diagram, we realize that the gain crossover frequency will be shifted to the right. We must offset the increased phase lag of $G_1(j\omega)$ due to this increase in the gain crossover frequency. Considering the shift of the gain crossover frequency, we may assume that $\phi_m$, the maximum phase lead required, is approximately 38°. (This means that 5° has been added to compensate for the shift in the gain crossover frequency.)

Since

$$\sin \phi_m = \frac{1 - \alpha}{1 + \alpha}$$

$\phi_m = 38°$ corresponds to $\alpha = 0.24$. Once the attenuation factor $\alpha$ has been determined on the basis of the required phase lead angle, the next step is to determine the corner frequencies $\omega = 1/T$ and $\omega = 1/(\alpha T)$ of the lead compensator. To do so, we first note that the maximum phase lead angle $\phi_m$ occurs at the geometric mean of the two corner frequencies, or $\omega = 1/(\sqrt{\alpha}T)$. [See Equation (9–2).] The amount of the modification in the magnitude curve at $\omega = 1/(\sqrt{\alpha}T)$ due to the inclusion of the term $(Ts + 1)/(\alpha Ts + 1)$ is

$$\left|\frac{1 + j\omega T}{1 + j\omega\alpha T}\right|_{\omega = 1/(\sqrt{\alpha}T)} = \left|\frac{1 + j\dfrac{1}{\sqrt{\alpha}}}{1 + j\alpha\dfrac{1}{\sqrt{\alpha}}}\right| = \frac{1}{\sqrt{\alpha}}$$

Note that

$$\frac{1}{\sqrt{\alpha}} = \frac{1}{\sqrt{0.24}} = \frac{1}{0.49} = 6.2\ \text{dB}$$

and $|G_1(j\omega)| = -6.2$ dB corresponds to $\omega = 9$ rad/sec. We shall select this frequency to be the new gain crossover frequency $\omega_c$. Noting that this frequency corresponds to $1/(\sqrt{\alpha}T)$, or $\omega_c = 1/(\sqrt{\alpha}T)$, we obtain

$$\frac{1}{T} = \sqrt{\alpha}\omega_c = 4.41$$

and

$$\frac{1}{\alpha T} = \frac{\omega_c}{\sqrt{\alpha}} = 18.4$$

The lead compensator thus determined is

$$G_c(s) = K_c \frac{s + 4.41}{s + 18.4} = K_c \alpha \frac{0.227s + 1}{0.054s + 1}$$

where the value of $K_c$ is determined as

$$K_c = \frac{K}{\alpha} = \frac{10}{0.24} = 41.7$$

Thus, the transfer function of the compensator becomes

$$G_c(s) = 41.7 \frac{s + 4.41}{s + 18.4} = 10 \frac{0.227s + 1}{0.054s + 1}$$

Note that

$$\frac{G_c(s)}{K} G_1(s) = \frac{G_c(s)}{10} 10G(s) = G_c(s)G(s)$$

The magnitude curve and phase-angle curve for $G_c(j\omega)/10$ are shown in Figure 9–8. The compensated system has the following open-loop transfer function:

$$G_c(s)G(s) = 41.7 \frac{s + 4.41}{s + 18.4} \frac{4}{s(s + 2)}$$

The solid curves in Figure 9–8 show the magnitude curve and phase-angle curve for the compensated system. The lead compensator causes the gain crossover frequency to increase from 6.3 to 9 rad/sec. The increase in this frequency means an increase in bandwidth. This implies an increase in the speed of response. The phase and gain margins are seen to be approximately 50° and $+\infty$ dB, respectively. The compensated system shown in Figure 9–9 therefore meets both the steady-state and the relative-stability requirements.

Note that for type 1 systems, such as the system just considered, the value of the static velocity error constant $K_v$ is merely the value of the frequency corresponding to the intersection of the extension of the initial $-20$-dB/decade slope line and the 0-dB line, as shown in Figure 9–8.

Figure 9–10 shows the polar plots of the uncompensated system $G_1(j\omega) = 10G(j\omega)$ and the compensated system $G_c(j\omega)G(j\omega)$. From Figure 9–10, we see that the resonant frequency of the uncompensated system is about 6 rad/sec and that of the compensated system is about 7 rad/sec. (This also indicates that the bandwidth has been increased.)

From Figure 9–10, we find that the value of the resonant peak $M_r$ for the uncompensated system with $K = 10$ is 3. The value of $M_r$ for the compensated system is found to be 1.29. This clearly shows that the compensated system has improved relative stability. (Note that

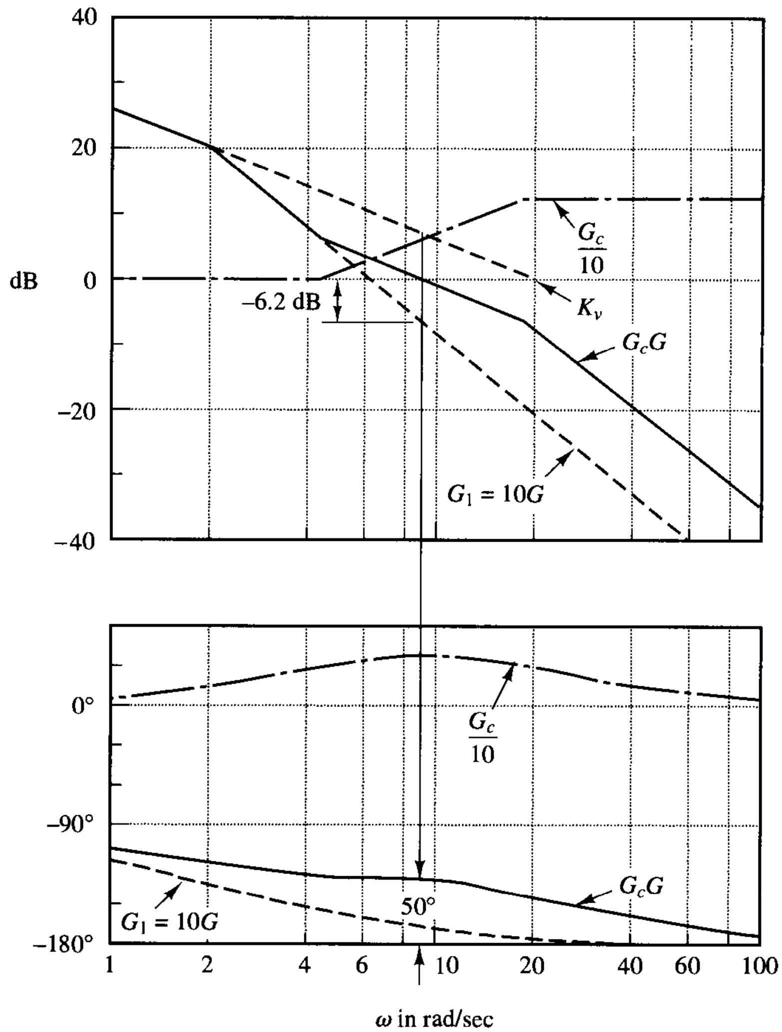



**Figure 9–8**
Bode diagram for the compensated system.

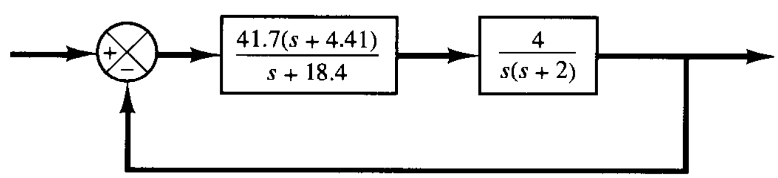



**Figure 9–9**
Compensated system.

the value of $M_r$ may be obtained easily by transferring the data from the Bode diagram to the Nichol's chart.)

Note that, if the phase angle of $G_1(j\omega)$ decreases rapidly near the gain crossover frequency, lead compensation becomes ineffective because the shift in the gain crossover frequency to the right makes it difficult to provide enough phase lead at the new gain crossover frequency. This means that, to provide the desired phase margin, we must use a very small value for $\alpha$. The value of $\alpha$, however, should not be too small (smaller than 0.05) nor should the maximum phase lead $\phi_m$ be too large (larger than 65°), because such values will require an additional gain of excessive value. [If more than 65° is needed, two (or more) lead networks may be used in series with an isolating amplifier.]

Finally, we shall examine the transient-response characteristics of the designed system. We shall obtain the unit-step response and unit-ramp response curves of the compensated and un-

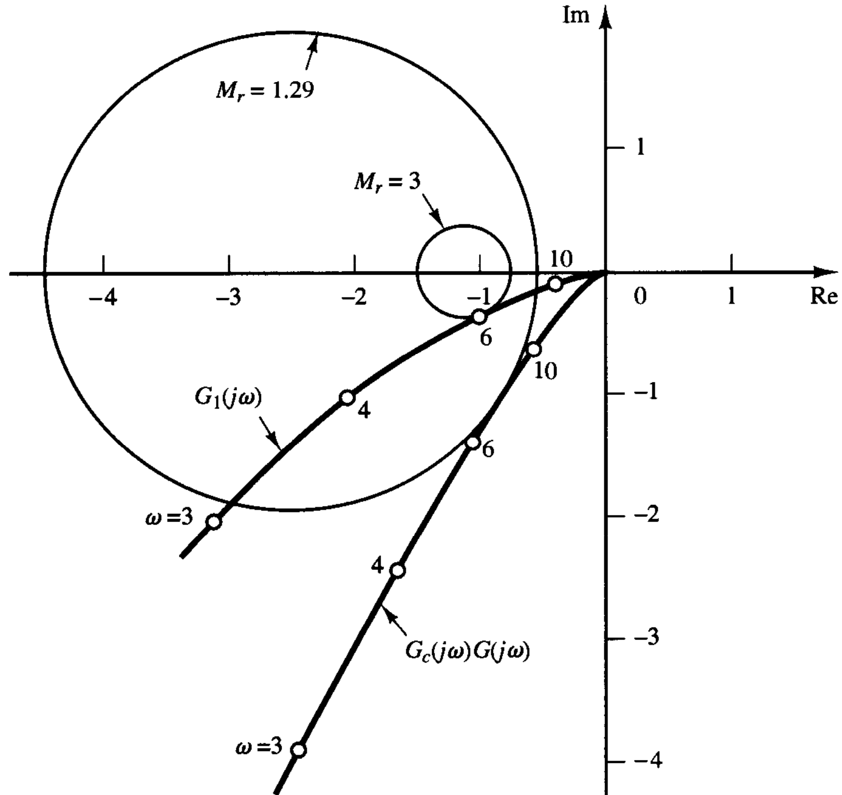



**Figure 9–10**
Polar plots of the uncompensated and compensated open-loop transfer function. ($G_1$: uncompensated system; $G_cG$: compensated system.)

compensated systems with MATLAB. Note that the closed-loop transfer functions of the uncompensated and compensated systems are given, respectively, by

$$\frac{C(s)}{R(s)} = \frac{4}{s^2 + 2s + 4}$$

and

$$\frac{C(s)}{R(s)} = \frac{166.8s + 735.588}{s^3 + 20.4s^2 + 203.6s + 735.588}$$

MATLAB programs for obtaining the unit-step response and unit-ramp response curves are given in MATLAB Program 9–1. Figure 9–11 shows the unit-step response curves and Figure 9–12 depicts the unit-ramp response curves. These response curves indicate that the designed system is satisfactory.

MATLAB Program 9–1

```
%*****Unit-step responses*****

num = [0   0   4];
den = [1   2   4];
numc = [0   0   166.8   735.588];
denc = [1   20.4   203.6   735.588];
t = 0:0.02:6;
```

```
[c1,x1,t] = step(num,den,t);
[c2,x2,t] = step(numc,denc,t);
plot [t,c1,'.',t,c2,'-')
grid
title('Unit-Step Responses of Compensated and Uncompensated Systems')
xlabel('t Sec')
ylabel('Outputs')
text(0.35,1.3,'Compensated system')
text(1.55,0.88,'Uncompensated system')

%*****Unit-ramp responses*****

num1 = [0   0   0   4];
den1 = [1   2   4   0];
num1c = [0   0   0   166.8   735.588];
den1c = [1   20.4   203.6   735.588   0];
t = 0:0.02:5;
[y1,z1,t] = step(num1,den1,t);
[y2,z2,t] = step(num1c,den1c,t);
plot(t,y1,'.',t,y2,'-',t,t,'--')
grid
title('Unit-Ramp Responses of Compensated and Uncompensated Systems')
xlabel('t Sec')
ylabel('Outputs')
text(0.89,3.7,'Compensated system')
text(2.25,1.1,'Uncompensated system')
```

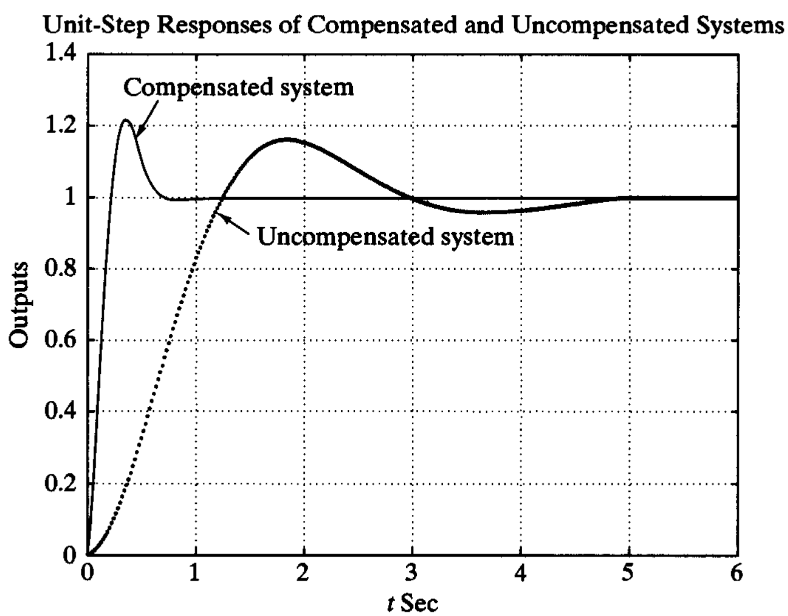



**Figure 9–11** Unit-step response curves of the compensated and uncompensated systems.

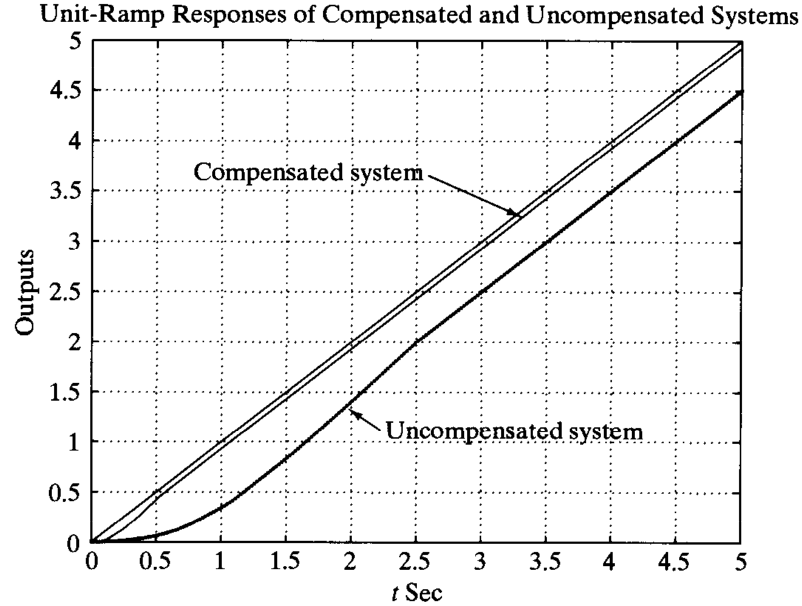



**Figure 9–12** Unit-ramp response curves of the compensated and uncompensated systems.

It is noted that the closed-loop poles for the compensated system are located as follows:

$$s = -6.9541 \pm j8.0592$$

$$s = -6.4918$$

Because the dominant closed-loop poles are located far from the $j\omega$ axis, the response damps out quickly.

## 9–3 LAG COMPENSATION

In this section we first discuss the Nyquist plot and Bode diagram of the lag compensator. Then we present lag compensation techniques based on the frequency-response approach.

**Characteristics of lag compensators.** Consider a lag compensator having the following transfer function:

$$G_c(s) = K_c\beta \frac{Ts + 1}{\beta Ts + 1} = K_c \frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}} \qquad (\beta > 1)$$

In the complex plane, a lag compensator has a zero at $s = -1/T$ and a pole at $s = -1/(\beta T)$. The pole is located to the right of the zero.

Figure 9–13 shows a polar plot of the lag compensator. Figure 9–14 shows a Bode diagram of the compensator, where $K_c = 1$ and $\beta = 10$. The corner frequencies of the lag compensator are at $\omega = 1/T$ and $\omega = 1/(\beta T)$. As seen from Figure 9–14, where the values of $K_c$ and $\beta$ are set equal to 1 and 10, respectively, the magnitude of the lag compensator becomes 10 (or 20 dB) at low frequencies and unity (or 0 dB) at high frequencies. Thus, the lag compensator is essentially a low-pass filter.

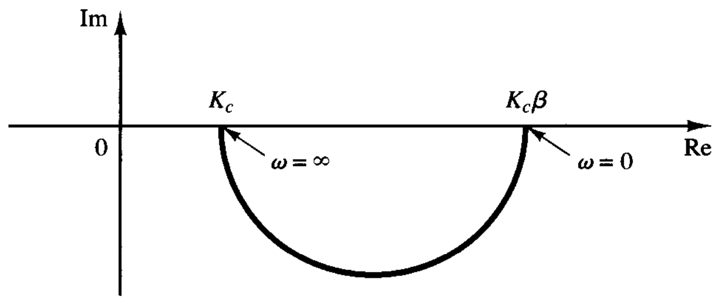



**Figure 9–13** Polar plot of a lag compensator $K_c\beta(j\omega T + 1)/(j\omega\beta T + 1)$.

**Lag compensation techniques based on the frequency-response approach.** The primary function of a lag compensator is to provide attenuation in the high-frequency range to give a system sufficient phase margin. The phase lag characteristic is of no consequence in lag compensation.

The procedure for designing lag compensators for the system shown in Figure 9–5 by the frequency-response approach may be stated as follows:

**1.** Assume the following lag compensator:

$$G_c(s) = K_c\beta \frac{Ts + 1}{\beta Ts + 1} = K_c \frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}} \qquad (\beta > 1)$$

Define

$$K_c\beta = K$$

Then

$$G_c(s) = K \frac{Ts + 1}{\beta Ts + 1}$$

The open-loop transfer function of the compensated system is

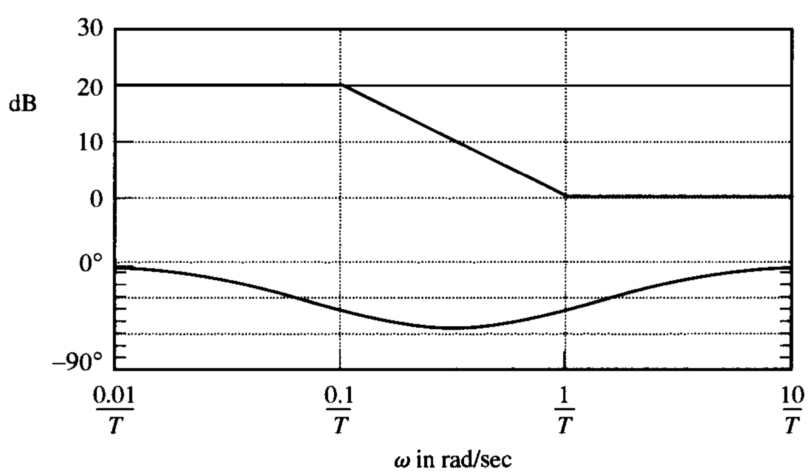



**Figure 9–14** Bode diagram of a lag compensator $\beta(j\omega T + 1)/(j\omega\beta T + 1)$, with $\beta = 10$.

$$G_c(s)G(s) = K\frac{Ts+1}{\beta Ts+1}G(s) = \frac{Ts+1}{\beta Ts+1}KG(s) = \frac{Ts+1}{\beta Ts+1}G_1(s)$$

where

$$G_1(s) = KG(s)$$

Determine gain $K$ to satisfy the requirement on the given static error constant.

**2.** If the uncompensated system $G_1(j\omega) = KG(j\omega)$ does not satisfy the specifications on the phase and gain margins, then find the frequency point where the phase angle of the open-loop transfer function is equal to $-180°$ plus the required phase margin. The required phase margin is the specified phase margin plus 5° to 12°. (The addition of 5° to 12° compensates for the phase lag of the lag compensator.) Choose this frequency as the new gain crossover frequency.

**3.** To prevent detrimental effects of phase lag due to the lag compensator, the pole and zero of the lag compensator must be located substantially lower than the new gain crossover frequency. Therefore, choose the corner frequency $\omega = 1/T$ (corresponding to the zero of the lag compensator) 1 octave to 1 decade below the new gain crossover frequency. (If the time constants of the lag compensator do not become too large, the corner frequency $\omega = 1/T$ may be chosen 1 decade below the new gain crossover frequency.)

**4.** Determine the attenuation necessary to bring the magnitude curve down to 0 dB at the new gain crossover frequency. Noting that this attenuation is $-20 \log \beta$, determine the value of $\beta$. Then the other corner frequency (corresponding to the pole of the lag compensator) is determined from $\omega = 1/(\beta T)$.

**5.** Using the value of $K$ determined in step 1 and that of $\beta$ determined in step 5, calculate constant $K_c$ from

$$K_c = \frac{K}{\beta}$$

**EXAMPLE 9–2** Consider the system shown in Figure 9–15. The open-loop transfer function is given by

$$G(s) = \frac{1}{s(s+1)(0.5s+1)}$$

It is desired to compensate the system so that the static velocity error constant $K_v$ is 5 $\text{sec}^{-1}$, the phase margin is at least 40°, and the gain margin is at least 10 dB.

We shall use a lag compensator of the form

$$G_c(s) = K_c\beta\frac{Ts+1}{\beta Ts+1} = K_c\frac{s+\frac{1}{T}}{s+\frac{1}{\beta T}} \qquad (\beta > 1)$$

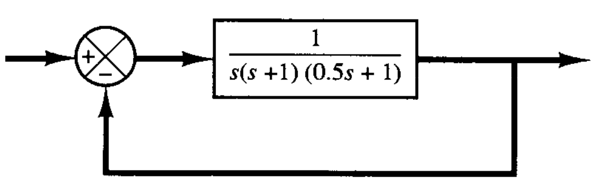



**Figure 9–15**
Control system.

Define

$$K_c\beta = K$$

Define also

$$G_1(s) = KG(s) = \frac{K}{s(s+1)(0.5s+1)}$$

The first step in the design is to adjust the gain $K$ to meet the required static velocity error constant. Thus,

$$K_v = \lim_{s\to 0} sG_c(s)G(s) = \lim_{s\to 0} s\frac{Ts+1}{\beta Ts+1}G_1(s) = \lim_{s\to 0} sG_1(s)$$

$$= \lim_{s\to 0} \frac{sK}{s(s+1)(0.5s+1)} = K = 5$$

or

$$K = 5$$

With $K = 5$, the compensated system satisfies the steady-state performance requirement.

We shall next plot the Bode diagram of

$$G_1(j\omega) = \frac{5}{j\omega(j\omega+1)(0.5j\omega+1)}$$

The magnitude curve and phase-angle curve of $G_1(j\omega)$ are shown in Figure 9–16. From this plot, the phase margin is found to be $-20°$, which means that the system is unstable.

Noting that the addition of a lag compensator modifies the phase curve of the Bode diagram, we must allow 5° to 12° to the specified phase margin to compensate for the modification of the phase curve. Since the frequency corresponding to a phase margin of 40° is 0.7 rad/sec, the new gain crossover frequency (of the compensated system) must be chosen near this value. To avoid overly large time constants for the lag compensator, we shall choose the corner frequency $\omega = 1/T$ (which corresponds to the zero of the lag compensator) to be 0.1 rad/sec. Since this corner frequency is not too far below the new gain crossover frequency, the modification in the phase curve may not be small. Hence, we add about 12° to the given phase margin as an allowance to account for the lag angle introduced by the lag compensator. The required phase margin is now 52°. The phase angle of the uncompensated open-loop transfer function is $-128°$ at about $\omega = 0.5$ rad/sec. So we choose the new gain crossover frequency to be 0.5 rad/sec. To bring the magnitude curve down to 0 dB at this new gain crossover frequency, the lag compensator must give the necessary attenuation, which in this case is $-20$ dB. Hence,

$$20 \log \frac{1}{\beta} = -20$$

or,

$$\beta = 10$$

The other corner frequency $\omega = 1(\beta T)$, which corresponds to the pole of the lag compensator, is then determined as

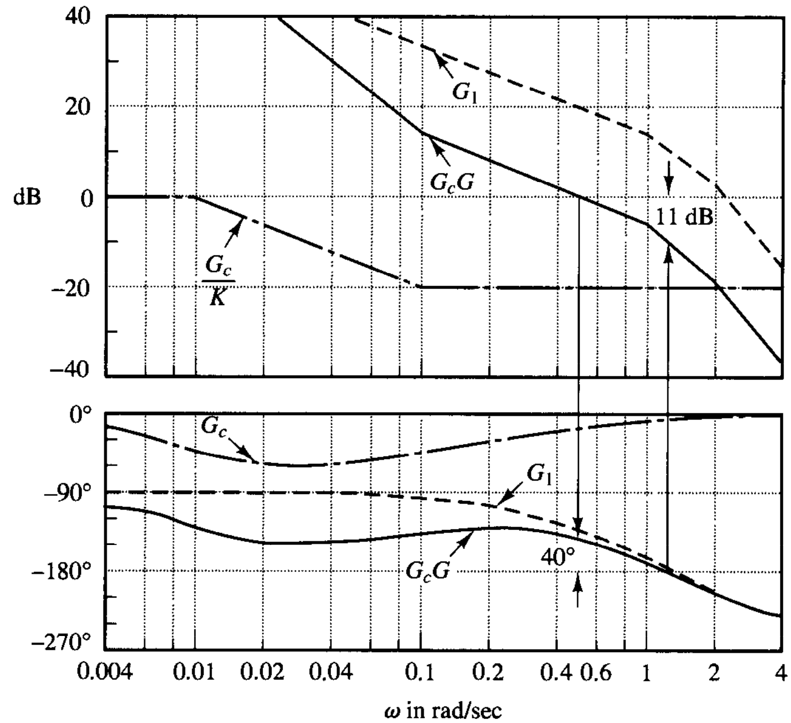



**Figure 9–16**
Bode diagrams for the uncompensated system, the compensator, and the compensated system. ($G_1$: uncompensated system, $G_c$: compensator, $G_cG$: compensated system.)

$$\frac{1}{\beta T} = 0.01 \text{ rad/sec}$$

Thus, the transfer function of the lag compensator is

$$G_c(s) = K_c(10)\frac{10s + 1}{100s + 1} = K_c\frac{s + \dfrac{1}{10}}{s + \dfrac{1}{100}}$$

Since the gain $K$ was determined to be 5 and $\beta$ was determined to be 10, we have

$$K_c = \frac{K}{\beta} = \frac{5}{10} = 0.5$$

The open-loop transfer function of the compensated system is

$$G_c(s)G(s) = \frac{5(10s + 1)}{s(100s + 1)(s + 1)(0.5s + 1)}$$

The magnitude and phase-angle curves of $G_c(j\omega)G(j\omega)$ are also shown in Figure 9–16.

The phase margin of the compensated system is about 40°, which is the required value. The gain margin is about 11 dB, which is quite acceptable. The static velocity error constant is 5 $\text{sec}^{-1}$, as required. The compensated system, therefore, satisfies the requirements on both the steady state and the relative stability.

Note that the new gain crossover frequency is decreased from approximately 1 to 0.5 rad/sec. This means that the bandwidth of the system is reduced.

To further show the effects of lag compensation, the log-magnitude versus phase plots of the gain-adjusted but uncompensated system $G_1(j\omega)$ and of the compensated system $G_c(j\omega)G(j\omega)$ are shown in Figure 9–17. The plot of $G_1(j\omega)$ clearly shows that the gain-adjusted but uncompensated system is unstable. The addition of the lag compensator stabilizes the system. The plot of $G_c(j\omega)G(j\omega)$ is tangent to the $M = 3$ dB locus. Thus, the resonant peak value is 3 dB, or 1.4, and this peak occurs at $\omega = 0.5$ rad/sec.

Compensators designed by different methods or by different designers (even using the same approach) may look sufficiently different. Any of the well-designed systems, however, will give similar transient and steady-state performance. The best among many alternatives may be chosen from the economic consideration that the time constants of the lag compensator should not be too large.

Finally, we shall examine the unit-step response and unit-ramp response of the compensated system and the original uncompensated system. The closed-loop transfer functions of the compensated and uncompensated systems are

$$\frac{C(s)}{R(s)} = \frac{50s + 5}{50s^4 + 150.5s^3 + 101.5s^2 + 51s + 5}$$

and

$$\frac{C(s)}{R(s)} = \frac{1}{0.5s^3 + 1.5s^2 + s + 1}$$

respectively. MATLAB Program 9–2 will produce the unit-step and unit-ramp responses of the compensated and uncompensated systems. The resulting unit-step response curves and unit-ramp response curves are shown in Figures 9–18 and 9–19, respectively. From the response curves we find that the designed system satisfies the given specifications and is satisfactory.

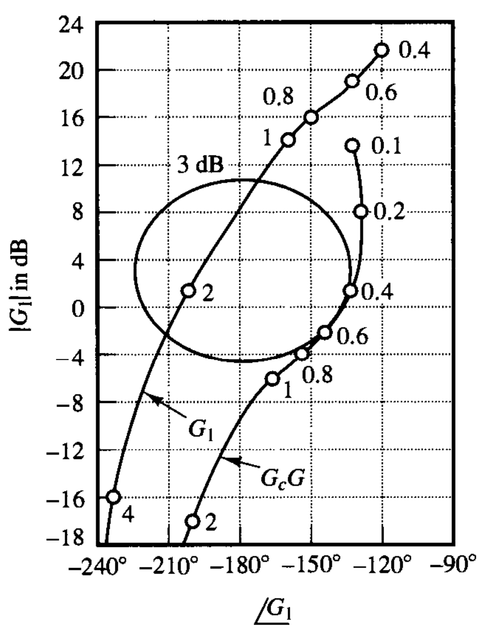



**Figure 9–17**
Log-magnitude versus phase plots of the uncompensated system and the compensated system. ($G_1$: uncompensated system, $G_cG$: compensated system.)

MATLAB Program 9–2

```
%*****Unit-step response*****

num = [0  0  0  1];
den = [0.5  1.5  1  1];
numc = [0  0  0  50  5];
denc = [50  150.5  101.5  51  5];
t = 0:0.1:40;
[c1,x1,t] = step(num,den,t);
[c2,x2,t] = step(numc,denc,t);
plot(t,c1,'.',t,c2,'-')
grid
title('Unit-Step Responses of Compensated and Uncompensated Systems')
xlabel('t Sec')
ylabel('Outputs')
text(12.2,1.27,'Compensated system')
text(12.2,0.7,'Uncompensated system')

%*****Unit-ramp response*****

num1 = [0  0  0  0  1];
den1 = [0.5  1.5  1  1  0];
num1c = [0  0  0  0  50  5];
den1c = [50  150.5  101.5  51  5  0];
t = 0:0.1:20;
[y1,z1,t] = step(num1,den1,t);
[y2,z2,t] = step(num1c,den1c,t);
plot(t,y1,'.',t,y2,'-',t,t,'--');
grid
title('Unit-Ramp Responses of Compensated and Uncompensated Systems')
xlabel('t Sec')
ylabel('Outputs')
text(8.4,3,'Compensated system')
text(8.4,5,'Uncompensated system')
```

Note that the zero and poles of the designed closed-loop systems are as follows:

$$\text{Zero at } s = -0.1$$

$$\text{Poles at } s = -0.2859 \pm j0.5196, \qquad s = -0.1228, \qquad s = -2.3155$$

The dominant closed-loop poles are very close to the $j\omega$ axis with the result that the response is slow. Also, a pair of the closed-loop pole at $s = -0.1228$ and the zero at $s = -0.1$ produces a slowly decreasing tail of small amplitude.

### A few comments on lag compensation

**1.** Lag compensators are essentially low-pass filters. Therefore, lag compensation permits a high gain at low frequencies (which improves the steady-state performance)

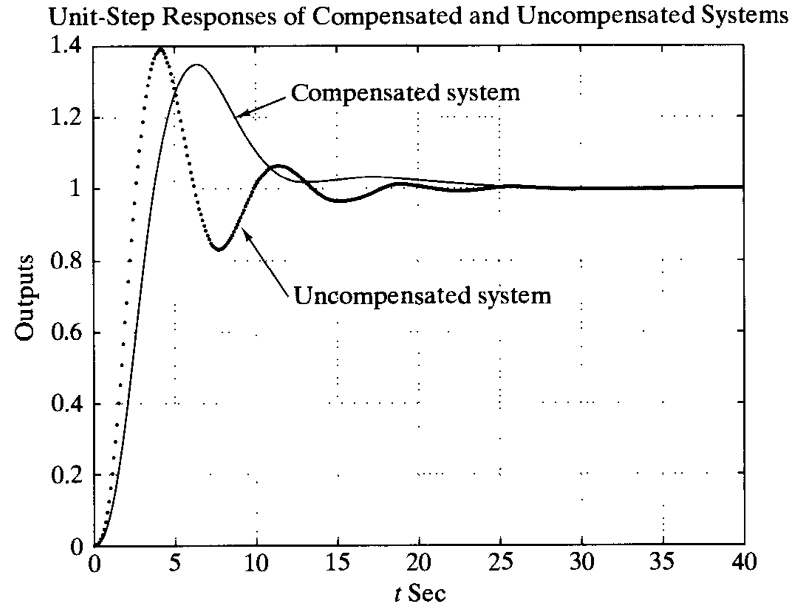



**Figure 9–18**
Unit-step response curves for the compensated and uncompensated systems (Example 9–2).

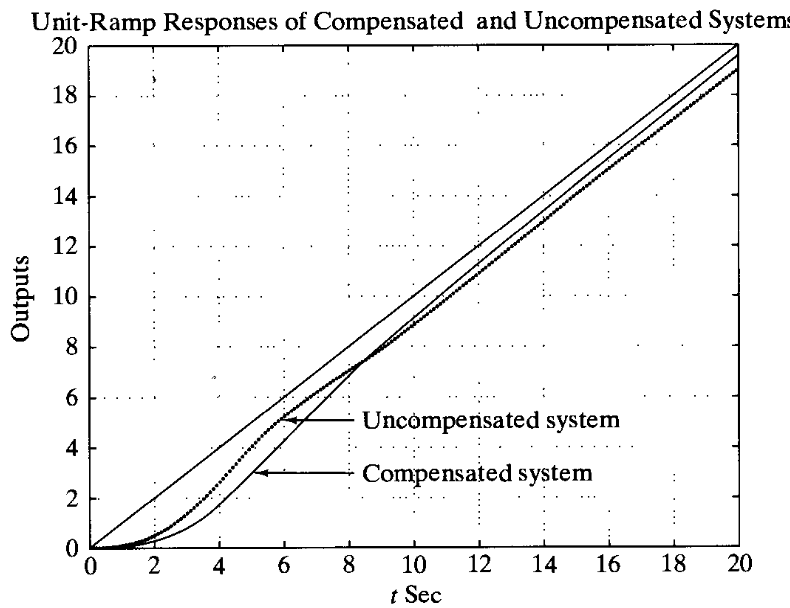



**Figure 9–19**
Unit-ramp response curves for the compensated and uncompensated systems (Example 9–2).

and reduces gain in the higher critical range of frequencies so as to improve the phase margin. Note that in lag compensation we utilize the attenuation characteristic of the lag compensator at high frequencies rather than the phase-lag characteristic. (The phase-lag characteristic is of no use for compensation purposes.)

**2.** Suppose that the zero and pole of a lag compensator are located at $s = -z$ and $s = -p$, respectively. Then the exact location of the zero and pole is not critical provided that they are close to the origin and the ratio $z/p$ is equal to the required multiplication factor of the static velocity error constant.

It should be noted, however, that the zero and pole of the lag compensator should not be located unnecessarily close to the origin, because the lag compensator will create an additional closed-loop pole in the same region as the zero and pole of the lag compensator.

The closed-loop pole located near the origin gives a very slowly decaying transient response, although its magnitude will become very small because the zero of the lag compensator will almost cancel the effect of this pole. However, the transient response (decay) due to this pole is so slow that the settling time will be adversely affected.

It is also noted that in the system compensated by a lag compensator the transfer function between the plant disturbance and the system error may not involve a zero that is near this pole. Therefore, the transient response to the disturbance input may last very long.

**3.** The attenuation due to the lag compensator will shift the gain crossover frequency to a lower frequency point where the phase margin is acceptable. Thus, the lag compensator will reduce the bandwidth of the system and will result in slower transient response. [The phase angle curve of $G_c(j\omega)G(j\omega)$ is relatively unchanged near and above the new gain crossover frequency.]

**4.** Since the lag compensator tends to integrate the input signal, it acts approximately as a proportional-plus-integral controller. Because of this, a lag-compensated system tends to become less stable. To avoid this undesirable feature, the time constant $T$ should be made sufficiently larger than the largest time constant of the system.

**5.** Conditional stability may occur when a system having saturation or limiting is compensated by use of a lag compensator. When the saturation or limiting takes place in the system, it reduces the effective loop gain. Then the system becomes less stable and unstable operation may even result, as shown in Figure 9–20. To avoid this, the system must be designed so that the effect of lag compensation becomes significant only when

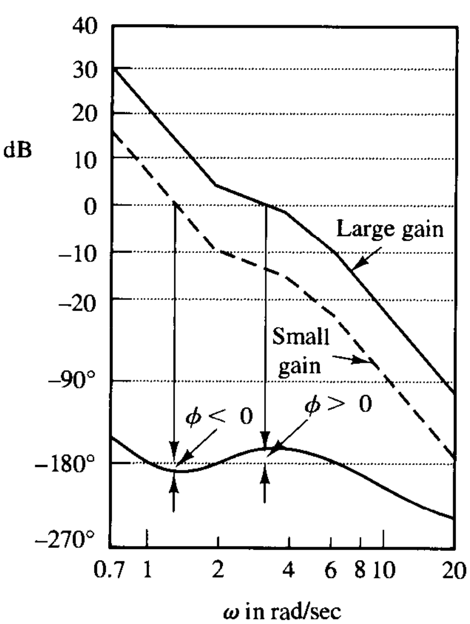



**Figure 9–20**
Bode diagram of a conditionally stable system.

the amplitude of the input to the saturating element is small. (This can be done by means of minor feedback-loop compensation.)

## 9–4 LAG–LEAD COMPENSATION

We shall first examine the frequency-response characteristics of the lag–lead compensator. Then we present the lag–lead compensation technique based on the frequency-response approach.

**Characteristic of lag–lead compensator.** Consider the lag–lead compensator given by

$$G_c(s) = K_c \left( \frac{s + \dfrac{1}{T_1}}{s + \dfrac{\gamma}{T_1}} \right) \left( \frac{s + \dfrac{1}{T_2}}{s + \dfrac{1}{\beta T_2}} \right) \tag{9–3}$$

where $\gamma > 1$ and $\beta > 1$. The term

$$\frac{s + \dfrac{1}{T_1}}{s + \dfrac{\gamma}{T_1}} = \frac{1}{\gamma} \left( \frac{T_1 s + 1}{\dfrac{T_1}{\gamma} s + 1} \right) \qquad (\gamma > 1)$$

produces the effect of the lead network, and the term

$$\frac{s + \dfrac{1}{T_2}}{s + \dfrac{1}{\beta T_2}} = \beta \left( \frac{T_2 s + 1}{\beta T_2 s + 1} \right) \qquad (\beta > 1)$$

produces the effect of the lag network.

In designing a lag–lead compensator, we frequently chose $\gamma = \beta$. (This is not necessary. We can, of course, choose $\gamma \neq \beta$.) In what follows, we shall consider the case where $\gamma = \beta$. The polar plot of the lag–lead compensator with $K_c = 1$ and $\gamma = \beta$ becomes as shown in Figure 9–21. It can be seen that, for $0 < \omega < \omega_1$, the compensator acts as a lag

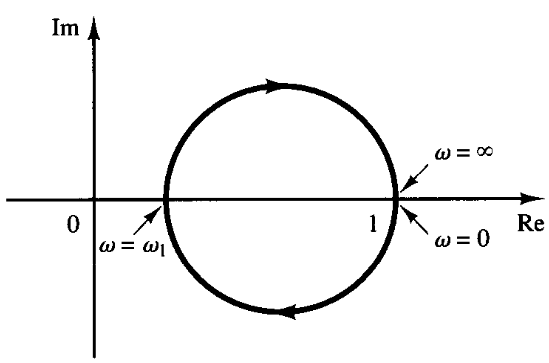



**Figure 9–21**
Polar plot of a lag–lead compensator given by Equation (9–3), with $K_c = 1$ and $\gamma = \beta$.

compensator, while for $\omega_1 < \omega < \infty$, it acts as a lead compensator. The frequency $\omega_1$ is the frequency at which the phase angle is zero. It is given by

$$\omega_1 = \frac{1}{\sqrt{T_1 T_2}}$$

(To derive this equation, see Problem A–9–2.)

Figure 9–22 shows the Bode diagram of a lag–lead compensator when $K_c = 1$, $\gamma = \beta = 10$, and $T_2 = 10T_1$. Notice that the magnitude curve has the value 0 dB at the low- and high-frequency regions.

**Lag–lead compensation based on the frequency-response approach.** The design of a lag–lead compensator by the frequency-response approach is based on the combination of the design techniques discussed under lead compensation and lag compensation.

Let us assume that the lag–lead compensator is of the following form:

$$G_c(s) = K_c \frac{(T_1 s + 1)(T_2 s + 1)}{\left(\dfrac{T_1}{\beta} s + 1\right)(\beta T_2 s + 1)} = K_c \frac{\left(s + \dfrac{1}{T_1}\right)\left(s + \dfrac{1}{T_2}\right)}{\left(s + \dfrac{\beta}{T_1}\right)\left(s + \dfrac{1}{\beta T_2}\right)} \qquad (9–4)$$

where $\beta > 1$. The phase lead portion of the lag–lead compensator (the portion involving $T_1$) alters the frequency-response curve by adding phase lead angle and increasing the phase margin at the gain crossover frequency. The phase lag portion (the portion involving $T_2$) provides attenuation near and above the gain crossover frequency and thereby allows an increase of gain at the low-frequency range to improve the steady-state performance.

We shall illustrate the details of the procedures for designing a lag–lead compensator by an example.

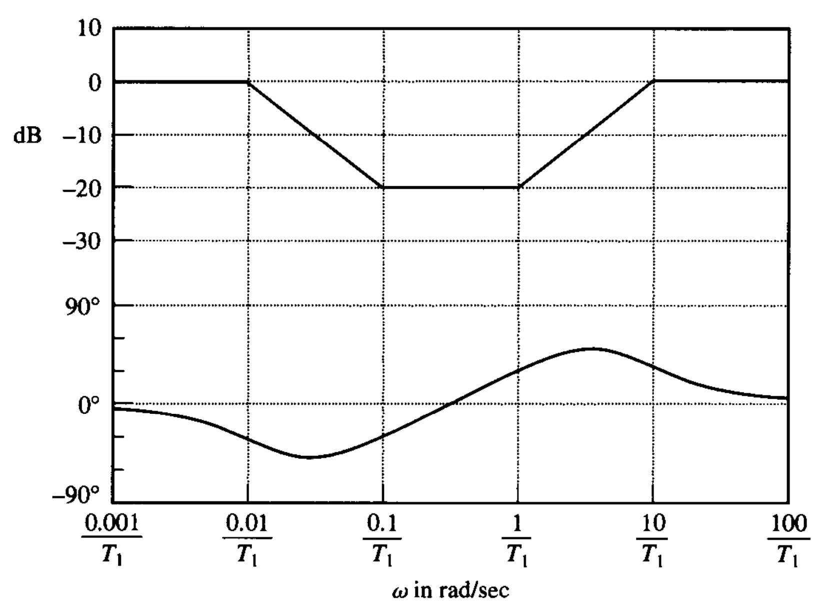



**Figure 9–22** Bode diagram of a lag–lead compensator given by Equation (9–3) with $K_c = 1, \gamma = \beta = 10$, and $T_2 = 10T_1$.

**EXAMPLE 9–3** Consider the unity-feedback system whose open-loop transfer function is

$$G(s) = \frac{K}{s(s+1)(s+2)}$$

It is desired that the static velocity error constant be 10 $\sec^{-1}$, the phase margin be 50°, and the gain margin be 10 dB or more.

Assume that we use the lag–lead compensator given by Equation (9–4). The open-loop transfer function of the compensated system is $G_c(s)G(s)$. Since the gain $K$ of the plant is adjustable, let us assume that $K_c = 1$. Then, $\lim_{s\to 0} G_c(s) = 1$.

From the requirement on the static velocity error constant, we obtain

$$K_v = \lim_{s\to 0} sG_c(s)G(s) = \lim_{s\to 0} sG_c(s)\frac{K}{s(s+1)(s+2)} = \frac{K}{2} = 10$$

Hence,

$$K = 20$$

We shall next draw the Bode diagram of the uncompensated system with $K = 20$, as shown in Figure 9–23. The phase margin of the uncompensated system is found to be −32°, which indicates that the uncompensated system is unstable.

The next step in the design of a lag–lead compensator is to choose a new gain crossover frequency. From the phase angle curve for $G(j\omega)$, we notice that $\underline{/G(j\omega)} = -180°$ at $\omega = 1.5$ rad/sec. It is convenient to choose the new gain crossover frequency to be 1.5 rad/sec so that the phase-lead angle required at $\omega = 1.5$ rad/sec is about 50°, which is quite possible by use of a single lag–lead network.

Once we choose the gain crossover frequency to be 1.5 rad/sec, we can determine the corner frequency of the phase lag portion of the lag–lead compensator. Let us choose the corner frequency $\omega = 1/T_2$ (which corresponds to the zero of the phase-lag portion of the compensator) to be 1 decade below the new gain crossover frequency, or at $\omega = 0.15$ rad/sec.

Recall that for the lead compensator the maximum phase lead angle $\phi_m$ is given by Equation (9–1), where $\alpha$ in Equation (9–1) is $1/\beta$ in the present case. By substituting $\alpha = 1/\beta$ in Equation (9–1), we have

$$\sin\phi_m = \frac{1 - \dfrac{1}{\beta}}{1 + \dfrac{1}{\beta}} = \frac{\beta - 1}{\beta + 1}$$

Notice that $\beta = 10$ corresponds to $\phi_m = 54.9°$. Since we need a 50° phase margin, we may choose $\beta = 10$. (Note that we will be using several degrees less than the maximum angle, 54.9°.) Thus,

$$\beta = 10$$

Then the corner frequency $\omega = 1/\beta T_2$ (which corresponds to the pole of the phase lag portion of the compensator) becomes $\omega = 0.015$ rad/sec. The transfer function of the phase lag portion of the lag–lead compensator then becomes

$$\frac{s + 0.15}{s + 0.015} = 10\left(\frac{6.67s + 1}{66.7s + 1}\right)$$

The phase lead portion can be determined as follows: Since the new gain crossover frequency is $\omega = 1.5$ rad/sec, from Figure 9–23, $G(j1.5)$ is found to be 13 dB. Hence, if the lag–lead compen-

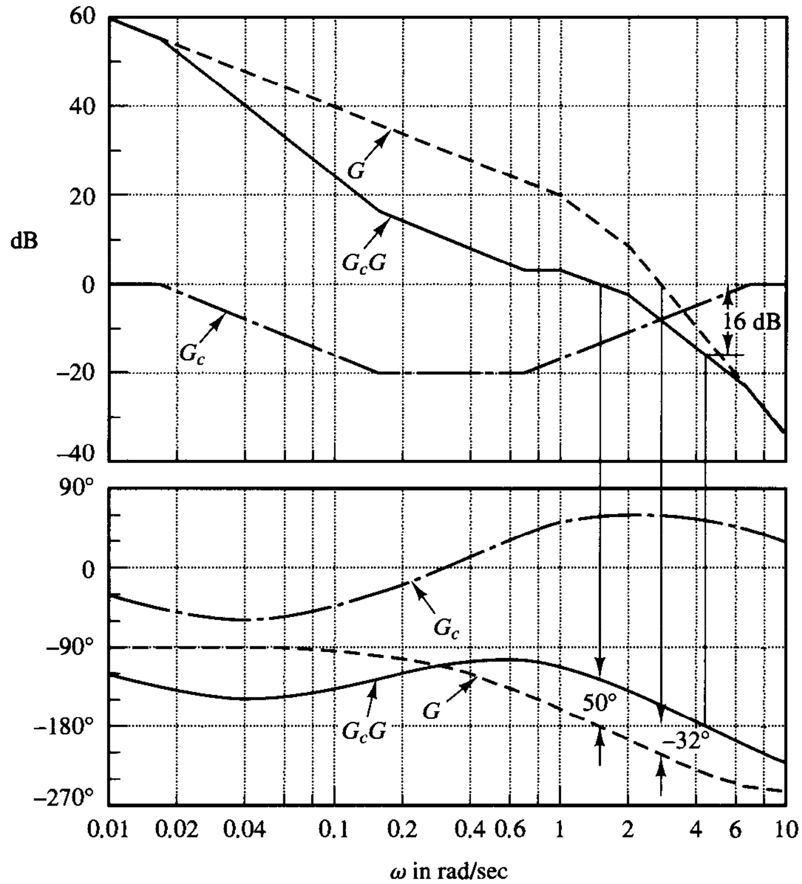



**Figure 9–23**
Bode diagrams for the uncompensated system, the compensator, and the compensated system. ($G$: uncompensated system, $G_c$: compensator, $G_cG$: compensated system.)

sator contributes $-13$ dB at $\omega = 1.5$ rad/sec, then the new gain crossover frequency is as desired. From this requirement, it is possible to draw a straight line of slope 20 dB/decade, passing through the point ($-13$ dB, 1.5 rad/sec). The intersections of this line and the 0-dB line and $-20$-dB line determine the corner frequencies. Thus, the corner frequencies for the lead portion are $\omega = 0.7$ rad/sec and $\omega = 7$ rad/sec. Thus, the transfer function of the lead portion of the lag–lead compensator becomes

$$\frac{s + 0.7}{s + 7} = \frac{1}{10}\left(\frac{1.43s + 1}{0.143s + 1}\right)$$

Combining the transfer functions of the lag and lead portions of the compensator, we obtain the transfer function of the lag–lead compensator. Since we chose $K_c = 1$, we have

$$G_c(s) = \left(\frac{s + 0.7}{s + 7}\right)\left(\frac{s + 0.15}{s + 0.015}\right) = \left(\frac{1.43s + 1}{0.143s + 1}\right)\left(\frac{6.67s + 1}{66.7s + 1}\right)$$

The magnitude and phase-angle curves of the lag–lead compensator just designed are shown in Figure 9–23. The open-loop transfer function of the compensated system is

$$G_c(s)G(s) = \frac{(s + 0.7)(s + 0.15)20}{(s + 7)(s + 0.015)s(s + 1)(s + 2)}$$

$$= \frac{10(1.43s + 1)(6.67s + 1)}{s(0.143s + 1)(66.7s + 1)(s + 1)(0.5s + 1)} \tag{9–5}$$

The magnitude and phase-angle curves of the system of Equation (9–5) are also shown in Figure 9–23. The phase margin of the compensated system is 50°, the gain margin is 16 dB, and the static velocity error constant is 10 $\text{sec}^{-1}$. All the requirements are therefore met, and the design has been completed.

Figure 9–24 shows the polar plots of the uncompensated system and compensated system. The $G_c(j\omega)G(j\omega)$ locus is tangent to the $M = 1.2$ circle at about $\omega = 2$ rad/sec. Clearly, this indicates that the compensated system has satisfactory relative stability. The bandwidth of the compensated system is slightly larger than 2 rad/sec.

In the following we shall examine the transient-response characteristics of the compensated system. (The uncompensated system is unstable.) The closed-loop transfer function of the compensated system is

$$\frac{C(s)}{R(s)} = \frac{95.381s^2 + 81s + 10}{4.7691s^5 + 47.7287s^4 + 110.3026s^3 + 163.724s^2 + 82s + 10}$$

The unit-step and unit-ramp response curves obtained with MATLAB are shown in Figures 9–25 and 9–26, respectively.

Note that the designed closed-loop control system has the following closed-loop zeros and poles:

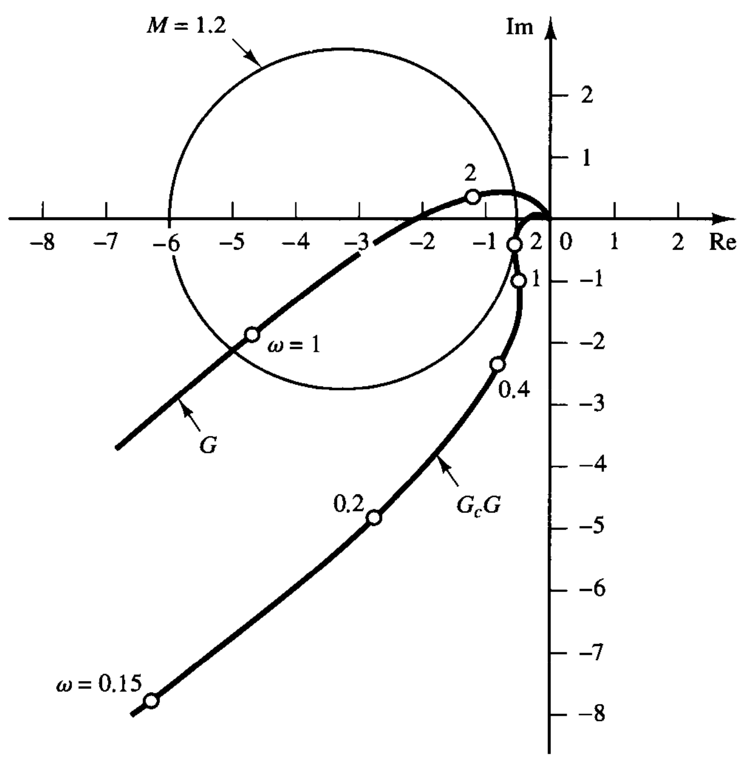



**Figure 9–24**
Polar plots of the uncompensated system and the compensated system. ($G$: uncompensated system, $G_cG$: compensated system.)

$$\text{Zeros at } s = -0.1499, \quad s = -0.6993$$

$$\text{Poles at } s = -0.8973 \pm j1.4439$$

$$s = -0.1785, \qquad s = -0.5425, \qquad s = -7.4923$$

The pole at $s = -0.1785$ and zero at $s = -0.1499$ are located very close to each other. Such a pair of pole and zero produces a long tail of small amplitude in the step response, as seen in Figure 9–25. Also, the pole at $s = -0.5425$ and zero at $s = -0.6993$ are located fairly close to each other. This pair adds an amplitude to the long tail.

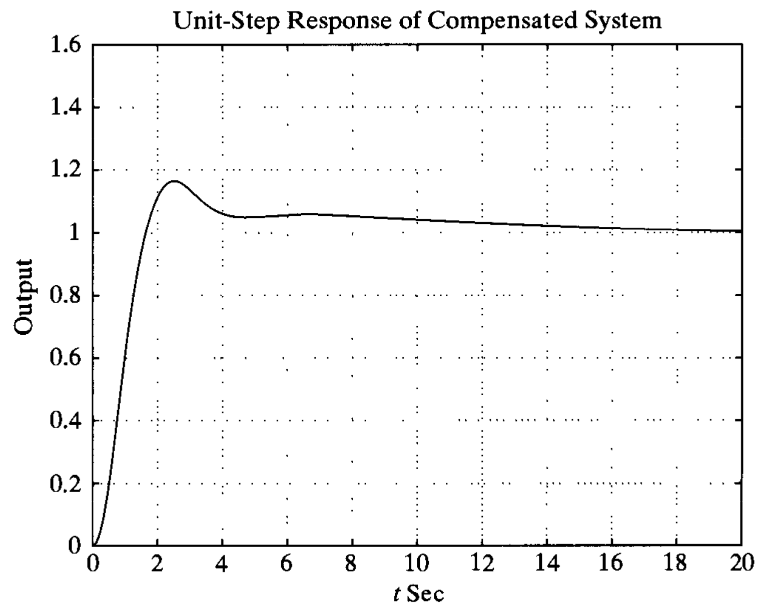



**Figure 9–25**
Unit-step response of the compensated system (Example 9–3).

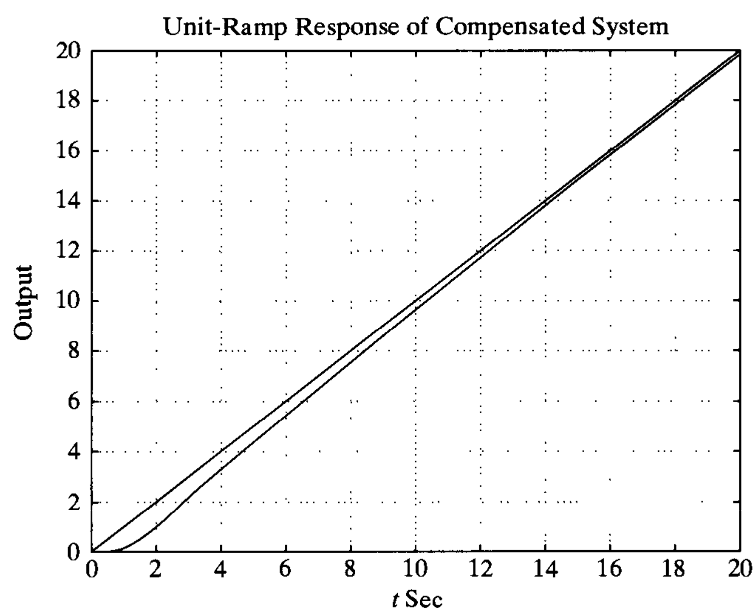



**Figure 9–26**
Unit-ramp response of the compensated system (Example 9–3).

## 9-5 CONCLUDING COMMENTS

This chapter has presented detailed procedures for designing lead, lag, and lag–lead compensators by the use of simple examples. We have shown that the design of a compensator to satisfy the given specifications (in terms of the phase margin and gain margin) can be carried out in the Bode diagram in a simple and straightforward manner. Note that a satisfactory design of a compensator for a complex system may require a creative application of these basic design principles.

### Comparison of lead, lag and lag–lead compensation

**1.** Lead compensation achieves the desired result through the merits of its phase-lead contribution, whereas lag compensation accomplishes the result through the merits of its attenuation property at high frequencies. (In some design problems both lag compensation and lead compensation may satisfy the specifications.)

**2.** Lead compensation is commonly used for improving stability margins. Lead compensation yields a higher gain crossover frequency than is possible with lag compensation. The higher gain crossover frequency means larger bandwidth. A large bandwidth means reduction in the settling time. The bandwidth of a system with lead compensation is always greater than that with lag compensation. Therefore, if a large bandwidth or fast response is desired, lead compensation should be employed. If, however, noise signals are present, then a large bandwidth may not be desirable, since it makes the system more susceptible to noise signals because of increase in the high-frequency gain.

**3.** Lead compensation requires an additional increase in gain to offset the attenuation inherent in the lead network. This means that lead compensation will require a larger gain than that required by lag compensation. A larger gain, in most cases, implies larger space, greater weight, and higher cost.

**4.** Lag compensation reduces the system gain at higher frequencies without reducing the system gain at lower frequencies. Since the system bandwidth is reduced, the system has a slower speed to respond. Because of the reduced high-frequency gain, the total system gain can be increased, and thereby low-frequency gain can be increased and the steady-state accuracy can be improved. Also, any high-frequency noises involved in the system can be attenuated.

**5.** If both fast responses and good static accuracy are desired, a lag–lead compensator may be employed. By use of the lag–lead compensator, the low-frequency gain can be increased (which means an improvement in steady-state accuracy), while at the same time the system bandwidth and stability margins can be increased.

**6.** Although a large number of practical compensation tasks can be accomplished with lead, lag, or lag–lead compensators, for complicated systems, simple compensation by use of these compensators may not yield satisfactory results. Then, different compensators having different pole–zero configurations must be employed.

**Graphical comparison.** Figure 9–27(a) shows a unit-step response curve and unit-ramp response curve of an uncompensated system. Typical unit-step response and unit-ramp response curves for the compensated system using a lead, lag, and lag–lead network, respectively, are shown in Figures 9–27(b), (c), and (d). The system with a lead

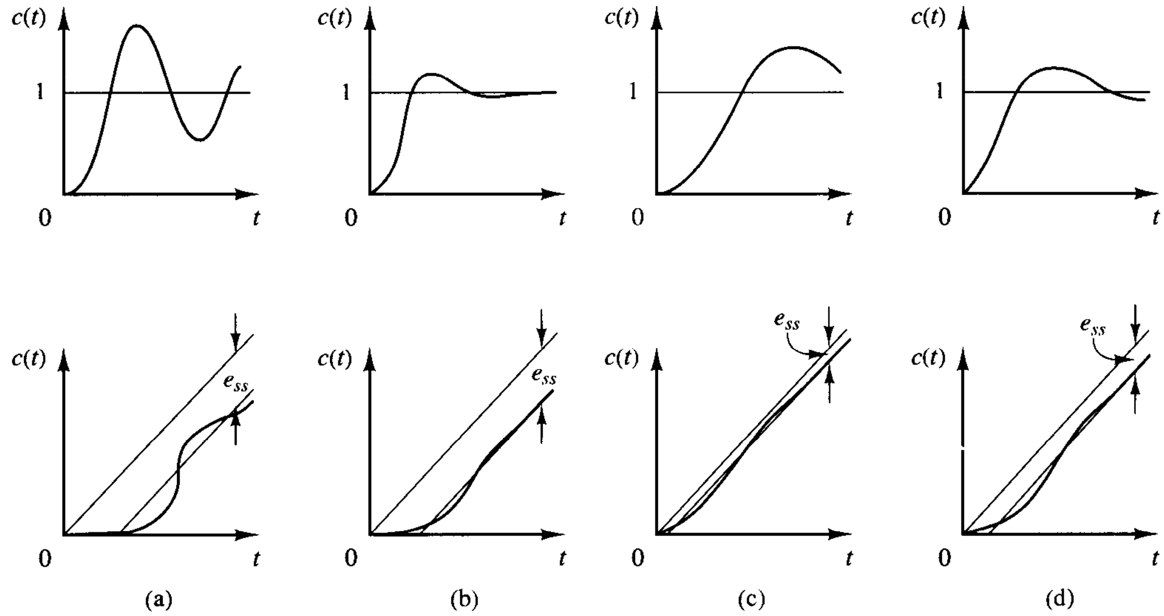



**Figure 9–27** Unit-step response curves and unit-ramp response curves. (a) Uncompensated system; (b) lead compensated system; (c) lag compensated system; (d) lag–lead compensated system.

compensator exhibits the fastest response, while that with a lag compensator exhibits the slowest response, but with marked improvements in the unit-ramp response. The system with a lag–lead compensator will give a compromise; reasonable improvements in both the transient response and steady-state response can be expected. The response curves shown depict the nature of improvements that may be expected from using different types of compensators.

**Feedback compensation.** A tachometer is one of the rate feedback devices. Another common rate feedback device is the rate gyro. Rate gyros are commonly used in aircraft autopilot systems.

Velocity feedback using a tachometer is very commonly used in positional servo systems. It is noted that, if the system is subjected to noise signals, velocity feedback may generate some difficulty if a particular velocity feedback scheme performs differentiation of the output signal. (The result is the accentuation of the noise effects.)

**Cancellation of undesirable poles.** Since the transfer function of elements in cascade is the product of their individual transfer functions, it is possible to cancel some undesirable poles or zeros by placing a compensating element in cascade, with its poles and zeros being adjusted to cancel the undesirable poles or zeros of the original system. For example, a large time constant $T_1$ may be canceled by use of the lead network $(T_1s + 1)/(T_2s + 1)$ as follows:

$$\left(\frac{1}{T_1s + 1}\right)\left(\frac{T_1s + 1}{T_2s + 1}\right) = \frac{1}{T_2s + 1}$$

If $T_2$ is much smaller than $T_1$, we can effectively eliminate the large time constant $T_1$. Figure 9–28 shows the effect of canceling a large time constant in step transient response.

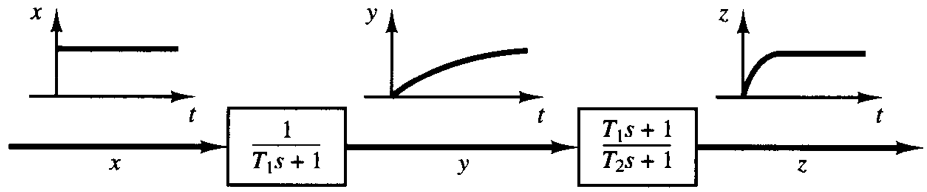



**Figure 9–28** Step-response curves showing the effect of canceling a large time constant.

If an undesirable pole in the original system lies in the right-half $s$ plane, this compensation scheme should not be used since, although mathematically it is possible to cancel the undesirable pole with an added zero, exact cancellation is physically impossible because of inaccuracies involved in the location of the poles and zeros. A pole in the right-half $s$ plane not exactly canceled by the compensator zero will eventually lead to unstable operation, because the response will involve an exponential term that increases with time.

It is noted that if a left-half plane pole is almost canceled but not exactly canceled, as is almost always the case, the uncanceled pole-zero combination will cause the response to have a small amplitude but long-lasting transient-response component. If the cancellation is not exact but is reasonably good, then this component will be quite small.

It should be noted that the ideal control system is not the one that has a transfer function of unity. Physically, such a control system cannot be built since it cannot instantaneously transfer energy from the input to the output. In addition, since noise is almost always present in one form or another, a system with a unity transfer function is not desirable. A desired control system, in many practical cases, may have one set of dominant complex-conjugate closed-loop poles with a reasonable damping ratio and undamped natural frequency. The determination of the significant part of the closed-loop pole-zero configuration, such as the location of the dominant closed-loop poles, is based on the specifications that give the required system performance.

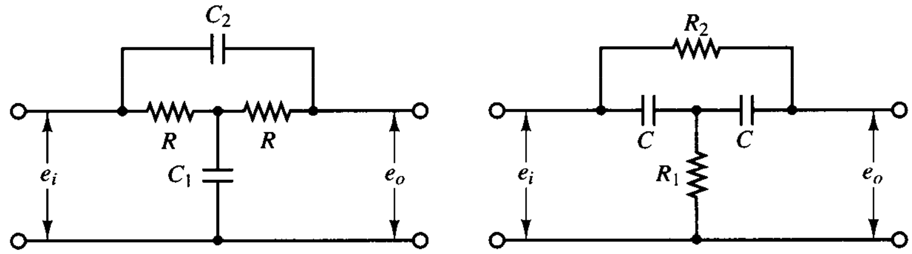
**Cancellation of undesirable complex-conjugate poles.** If the transfer function of a plant contains one or more pairs of complex-conjugate poles, then a lead, lag, or lag–lead compensator may not give satisfactory results. In such a case, a network that has two zeros and two poles may prove to be useful. If the zeros are chosen so as to cancel the undesirable complex-conjugate poles of the plant, then we can essentially replace the undesirable poles by acceptable poles. That is, if the undesirable complex-conjugate poles are in the left-half $s$ plane and are in the form

$$\frac{1}{s^2 + 2\zeta_1\omega_1 s + \omega_1^2}$$

then the insertion of a compensating network having the transfer function

$$\frac{s^2 + 2\zeta_1\omega_1 s + \omega_1^2}{s^2 + 2\zeta_2\omega_2 s + \omega_2^2}$$

will result in an effective change of the undesirable complex-conjugate poles to acceptable poles. Note that even though the cancellation may not be exact the compensated system will exhibit better response characteristics. (As stated earlier, this approach cannot be used if the undesirable complex-conjugate poles are in the right-half $s$ plane.)




$$\frac{E_o(s)}{E_i(s)} = \frac{RC_1RC_2s^2 + 2RC_2s + 1}{RC_1RC_2s^2 + (RC_1 + 2RC_2)s + 1}$$

(a)

$$\frac{E_o(s)}{E_i(s)} = \frac{R_1CR_2Cs^2 + 2R_1Cs + 1}{R_1CR_2Cs^2 + (R_2C + 2R_1C)s + 1}$$

(b)

**Figure 9–29**
Bridged-$T$ networks.

Familiar networks consisting only of $RC$ components whose transfer functions possess two zeros and two poles are the bridged-$T$ networks. Examples of bridged-$T$ networks and their transfer functions are shown in Figure 9–29.

**Concluding comments.** In the design examples presented in this chapter, we have been primarily concerned only with the transfer functions of compensators. In actual design problems, we must choose the hardware. Thus, we must satisfy additional design constraints such as cost, size, weight, and reliability.

The system designed may meet the specifications under normal operating conditions but may deviate considerably from the specifications when environmental changes are considerable. Since the changes in the environment affect the gain and time constants of the system, it is necessary to provide automatic or manual means to adjust the gain to compensate for such environmental changes, for nonlinear effects that were not taken into account in the design, and also to compensate for manufacturing tolerances from unit to unit in the production of system components. (The effects of manufacturing tolerances are suppressed in a closed-loop system; therefore, the effects may not be critical in closed-loop operation but critical in open-loop operation.) In addition to this, the designer must remember that any system is subject to small variations due mainly to the normal deterioration of the system.

## EXAMPLE PROBLEMS AND SOLUTIONS

**A-9-1.** Show that the lead network and lag network inserted in cascade in an open loop acts as proportional-plus-derivative control (in the region of small $\omega$) and proportional-plus-integral control (in the region of large $\omega$), respectively.

**Solution.** In the region of small $\omega$, the polar plot of the lead network is approximately the same as that of the proportional-plus-derivative controller. This is shown in Figure 9–30(a).

Similarly, in the region of large $\omega$, the polar plot of the lag network approximates the proportional-plus-integral controller, as shown in Figure 9–30(b).

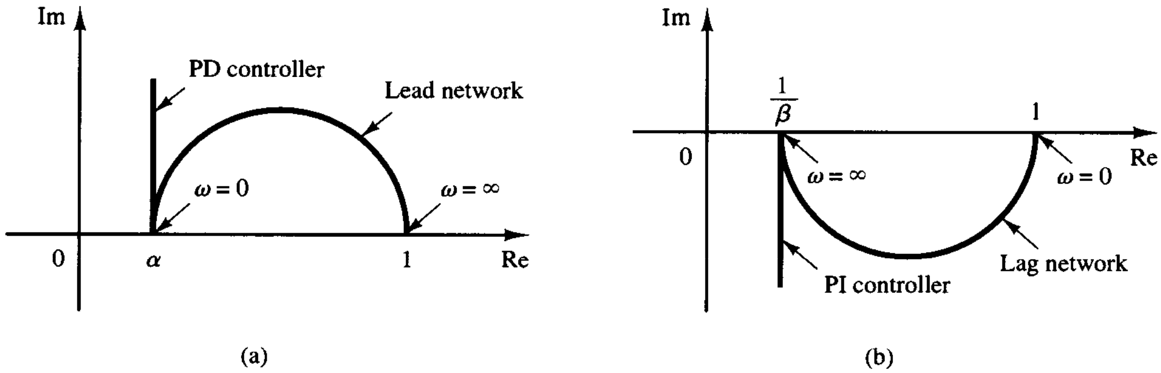



**Figure 9–30**
(a) Polar plots of a lead network and a proportional-plus-derivative controller; (b) polar plots of a lag network and a proportional-plus-integral controller.

**A-9-2.** Consider a lag–lead compensator $G_c(s)$ defined by

$$G_c(s) = K_c \frac{\left(s + \frac{1}{T_1}\right)\left(s + \frac{1}{T_2}\right)}{\left(s + \frac{\beta}{T_1}\right)\left(s + \frac{1}{\beta T_2}\right)}$$

Show that at frequency $\omega_1$, where

$$\omega_1 = \frac{1}{\sqrt{T_1 T_2}}$$

the phase angle of $G_c(j\omega)$ becomes zero. (This compensator acts as a lag compensator for $0 < \omega < \omega_1$ and acts as a lead compensator for $\omega_1 < \omega < \infty$.)

**Solution.** The angle of $G_c(j\omega)$ is given by

$$\begin{aligned}\underline{/G_c(j\omega)} &= \underline{/j\omega + \frac{1}{T_1}} + \underline{/j\omega + \frac{1}{T_2}} - \underline{/j\omega + \frac{\beta}{T_1}} - \underline{/j\omega + \frac{1}{\beta T_2}} \\ &= \tan^{-1}\omega T_1 + \tan^{-1}\omega T_2 - \tan^{-1}\omega T_1/\beta - \tan^{-1}\omega T_2 \beta\end{aligned}$$

At $\omega = \omega_1 = 1/\sqrt{T_1 T_2}$, we have

$$\underline{/G_c(j\omega_1)} = \tan^{-1}\sqrt{\frac{T_1}{T_2}} + \tan^{-1}\sqrt{\frac{T_2}{T_1}} - \tan^{-1}\frac{1}{\beta}\sqrt{\frac{T_1}{T_2}} - \tan^{-1}\beta\sqrt{\frac{T_2}{T_1}}$$

Since

$$\tan\left(\tan^{-1}\sqrt{\frac{T_1}{T_2}} + \tan^{-1}\sqrt{\frac{T_2}{T_1}}\right) = \frac{\sqrt{\frac{T_1}{T_2}} + \sqrt{\frac{T_2}{T_1}}}{1 - \sqrt{\frac{T_1}{T_2}}\sqrt{\frac{T_2}{T_1}}} = \infty$$

or

$$\tan^{-1}\sqrt{\frac{T_1}{T_2}} + \tan^{-1}\sqrt{\frac{T_2}{T_1}} = 90°$$

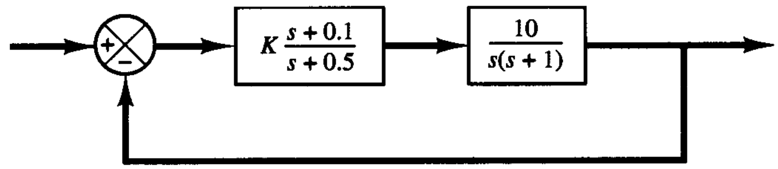



**Figure 9–31**
Control system.

and also

$$\tan^{-1}\frac{1}{\beta}\sqrt{\frac{T_1}{T_2}} + \tan^{-1}\beta\sqrt{\frac{T_2}{T_1}} = 90°$$

we have

$$\underline{/G_c(j\omega_1)} = 0°$$

Thus, the angle of $G_c(j\omega_1)$ becomes 0° at $\omega = \omega_1 = 1/\sqrt{T_1T_2}$.

**A-9-3.** Consider the control system shown in Figure 9–31. Determine the value of gain $K$ such that the phase margin is 60°.

**Solution.** The open-loop transfer function is

$$G(s) = K\frac{s+0.1}{s+0.5}\frac{10}{s(s+1)}$$
$$= \frac{K(10s+1)}{s^3 + 1.5s^2 + 0.5s}$$

Let us plot the Bode diagram of $G(s)$ when $K = 1$. MATLAB Program 9–3 may be used for this purpose. Figure 9–32 shows the Bode diagram produced by this program. From this diagram the required phase margin of 60° occurs at the frequency $\omega = 1.15$ rad/sec. The magnitude of $G(j\omega)$ at this frequency is found to be 14.5 dB. Then gain $K$ must satisfy the following equation:

$$20 \log K = -14.5 \text{ dB}$$

MATLAB Program 9–3

```
num = [0   0   10   1];
den = [1   1.5   0.5   0];
bode(num,den)
subplot(2,1,1);
title('Bode Diagram of G(s) = (10s + 1)/[s(s + 0.5)(s + 1)]')
```

or

$$K = 0.188$$

Thus, we have determined the value of gain $K$.

To verify the results, let us draw a Nyquist plot of $G$ for the frequency range

$$w = 0.1{:}0.01{:}1.15$$

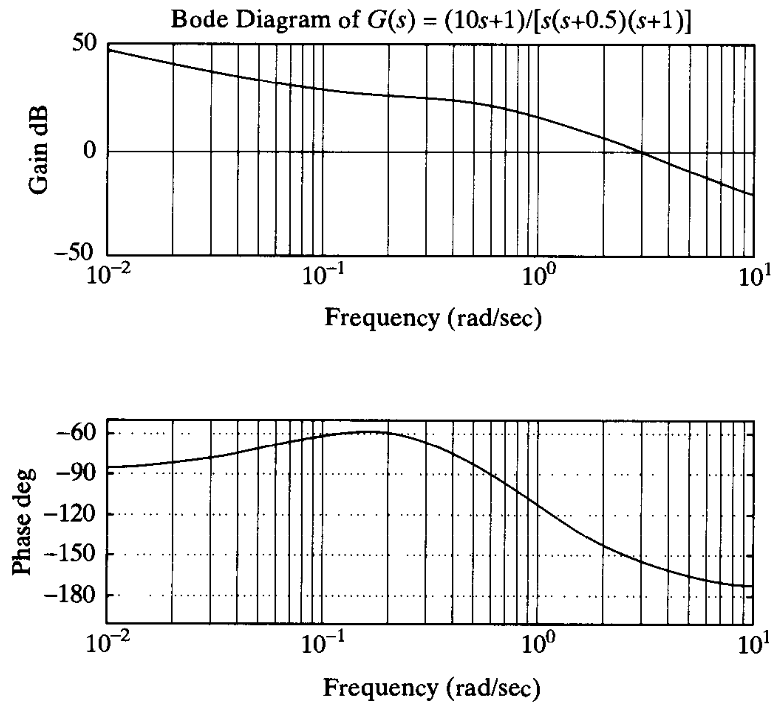



**Figure 9–32**
Bode diagram of $G(s) = \dfrac{10s + 1}{s(s + 0.5)(s + 1)}$.

The end point of the locus ($w = 1.15$ rad/sec) will be on a unit circle in the Nyquist plane. To check the phase margin, it is convenient to draw the Nyquist plot on a polar diagram, using polar grids.

To draw the Nyquist plot on a polar diagram, first define a complex vector $z$ by

```
z = re + i*im = re^iθ
```

where r and θ (theta) are given by

```
r = abs(z)
theta = angle(z)
```

The abs means the square root of the sum of the real part squared and imaginary part squared, angle means $\tan^{-1}$ (imaginary part/real part).

If we use the command

```
polar(theta,r)
```

MATLAB will produce a plot in the polar coordinates. Subsequent use of the *grid* command draws polar grid lines and grid circles.

MATLAB Program 9–4 produces the Nyquist plot of $G(j\omega)$, where $\omega$ is between 0.5 and 1.15 rad/sec. The resulting plot is shown in Figure 9–33. Notice that point $G(j1.15)$ lies on the unit circle, and the phase angle of this point is $-120°$. Hence, the phase margin is 60°. The fact that point $G(j1.15)$ is on the unit circle verifies that at $\omega = 1.15$ rad/sec the magnitude is equal to 1 or 0 dB.

MATLAB Program 9–4

```
%*****Nyquist plot in polar coordinates*****

num = [0   0   1.88   0.188];
den = [1   1.5   0.5   0];
w = 0.5:0.01:1.15;
[re,im,w] = nyquist(num,den,w);

%*****Convert rectangular coordinates into polar coordinates
% by defining z, r, theta as follows*****

z = re + i*im;
r = abs(z);
theta = angle(z);

%*****To draw polar plot, enter command 'polar(theta,r)'*****

polar(theta,r)
title('Check of Phase Margin')
text(0.1,-1.5,'Nyquist plot')
text(-2.25,-0.3,'Phase margin')
text(-2.25,-0.7,'is 60 degrees.')
```

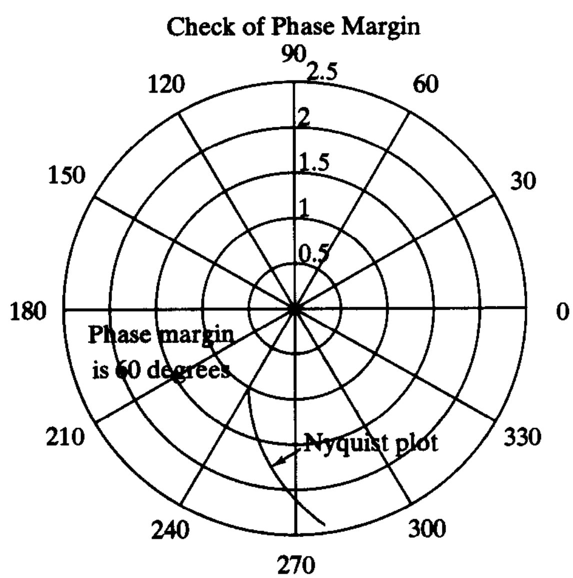



**Figure 9–33**
Nyquist plot of $G(j\omega)$ showing that the phase margin is 60°.

(Thus, $\omega = 1.15$ is the gain crossover frequency.) Thus, $K = 0.188$ gives the desired phase margin of 60°.

Note that in writing 'text' in the polar diagram we enter the *text* command as follows:

```
text(x,y,' ')
```

For example, to write 'Nyquist plot' starting at point (0.1, −1.5), enter the command

```
text(0.1, -1.5,'Nyquist plot')
```

The text is written horizontally on the screen.

**A-9-4.** If the open-loop transfer function $G(s)$ involves lightly damped complex-conjugant poles, then more than one $M$ locus may be tangent to the $G(j\omega)$ locus.

Consider the unity-feedback system whose open-loop transfer function is

$$G(s) = \frac{9}{s(s + 0.5)(s^2 + 0.6s + 10)} \tag{9–6}$$

Draw the Bode diagram for this open-loop transfer function. Draw also the log-magnitude versus phase plot, and show that two $M$ loci are tangent to the $G(j\omega)$ locus. Finally, plot the Bode diagram for the closed-loop transfer function.

**Solution.** Figure 9–34 shows the Bode diagram of $G(j\omega)$. Figure 9–35 shows the log-magnitude versus phase plot of $G(j\omega)$. It is seen that the $G(j\omega)$ locus is tangent to the $M$ = 8-dB locus at $\omega = 0.97$ rad/sec, and it is tangent to the $M$ = −4-dB locus at $\omega = 2.8$ rad/sec.

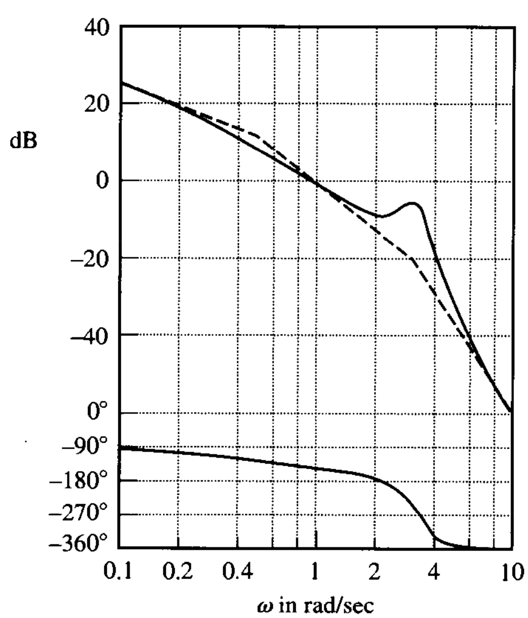



**Figure 9–34**
Bode diagram of $G(j\omega)$ given by Equation (9–6).

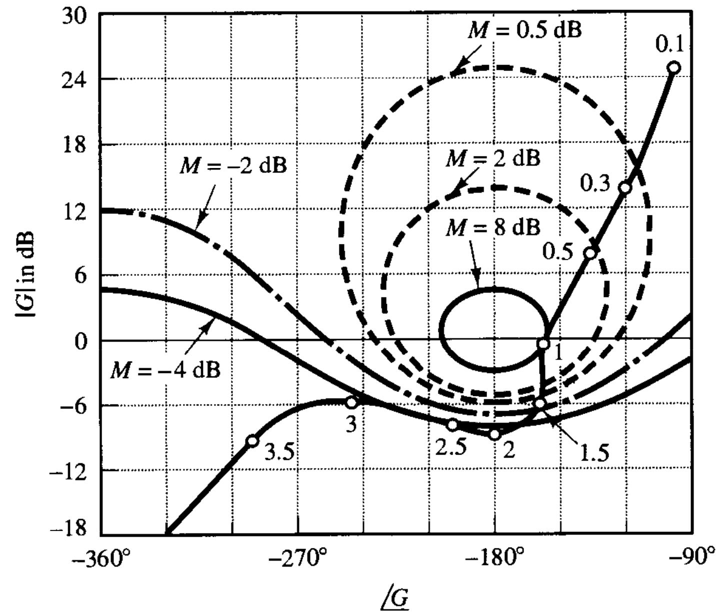



**Figure 9–35**
Log-magnitude versus phase plot of $G(j\omega)$ given by Equation (9–6).

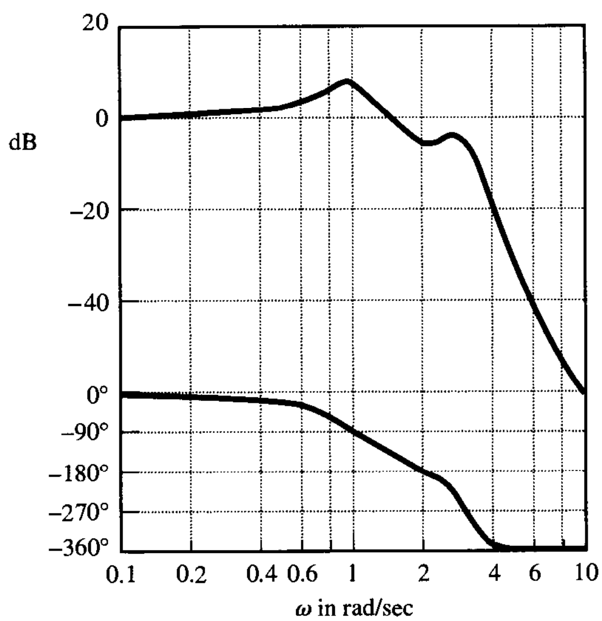



**Figure 9–36**
Bode diagram of $G(j\omega)/[1 + G(j\omega)]$, where $G(j\omega)$ is given by Equation (9–6).

Figure 9–36 shows the Bode diagram of the closed-loop transfer function. The magnitude curve of the closed-loop frequency response shows two resonant peaks. Note that such a case occurs when the closed-loop transfer function involves the product of two lightly damped second-order terms and the two corresponding resonant frequencies are sufficiently separated from each other. As a matter of fact, the closed-loop transfer function of this system can be written

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

$$= \frac{9}{(s^2 + 0.487s + 1)(s^2 + 0.613s + 9)}$$

Clearly, the closed-loop transfer function is a product of two lightly damped second-order terms (the damping ratios are 0.243 and 0.102), and the two resonant frequencies are sufficiently separated.

**A-9-5.** Consider a unity-feedback system whose feedforward transfer function is given by

$$G(s) = \frac{1}{s^2}$$

It is desired to insert a series compensator so that the open-loop frequency-response curve is tangent to the $M$ = 3-dB circle at $\omega$ = 3 rad/sec. The system is subjected to high-frequency noises and sharp cutoff is desired. Design an appropriate series compensator.

**Solution.** To stabilize the system, we may insert a proportional-plus-derivative type of compensator or a lead compensator. Since sharp cutoff is required, we choose a lead compensator. Consider the following lead compensator:

$$G_c(s) = K_c \frac{Ts + 1}{\alpha Ts + 1} \quad (\alpha < 1)$$

The compensated open-loop frequency-response curve must be tangent to the $M$ = 3-dB locus. To minimize the additional gain $K_c$, we choose the tangent point to the 3-dB locus as shown in Figure 9–37. From Figure 9–37 we see that the lead compensator must provide about 45°. Then the necessary value of $\alpha$ is determined from

$$\sin 45° = \frac{1 - \alpha}{1 + \alpha}$$

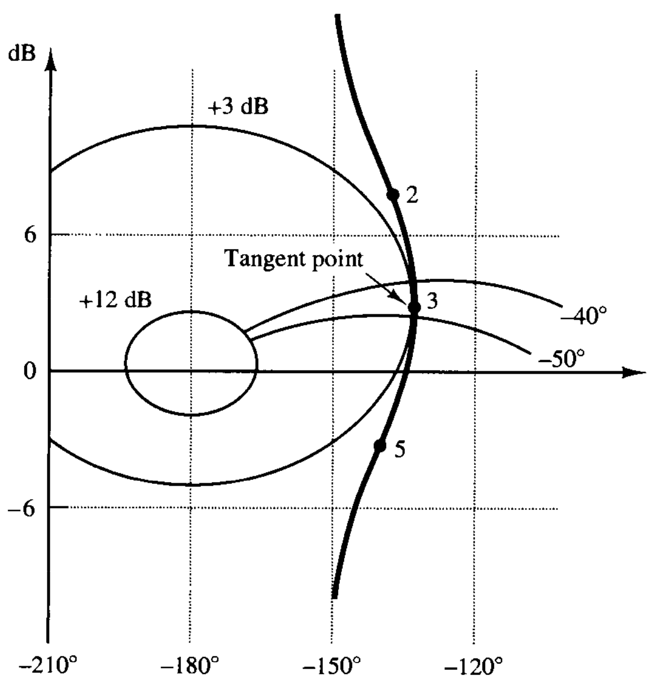



**Figure 9–37**
Nichols chart showing that the $G_c(j\omega)G(j\omega)$ locus is tangent to the $M$ = 3-dB locus at $\omega$ = 3 rad/sec.

or $\alpha = 0.172 \doteqdot \frac{1}{6}$. Let us choose $\alpha = \frac{1}{6}$. Since it is required that the open-loop frequency-response curve $G_c(j\omega)G(j\omega)$ be tangent to the $M$ = 3-dB locus at $\omega$ = 3 rad/sec, we obtain

$$20 \log |G_c(j\omega)G(j\omega)|_{\omega=3} = 20 \log |G_c(j3)| + 20 \log |G(j3)|$$

$$= 20 \log |G_c(j3)| + 20 \log \left|\frac{1}{9}\right| = 3 \text{ dB}$$

or

$$20 \log |G_c(j3)| = 22.085 \text{ dB}$$

The two time constants $T$ and $\alpha T$ of the lead compensator can be determined as follows: Noting that

$$\sqrt{\frac{1}{T} \cdot \frac{1}{\alpha T}} = 3$$

we have

$$\frac{1}{T} = \frac{3}{\sqrt{6}} = 1.225, \qquad \frac{1}{\alpha T} = 3\sqrt{6} = 7.348$$

From the Bode diagram as shown in Figure 9–38, we find the gain $K_c$ to be 14.3 dB or 5.19. Thus, the designed compensator is given by

$$G_c(s) = 5.19 \frac{0.816s + 1}{0.136s + 1}$$

**A-9-6.** Consider the system shown in Figure 9–39. Design a lead compensator such that the closed-loop system will have the phase margin of 50° and gain margin of not less than 10 dB. Assume that

$$G_c(s) = K_c\alpha\left(\frac{Ts + 1}{\alpha Ts + 1}\right) \qquad (0 < \alpha < 1)$$

It is desired that the bandwidth of the closed-loop system be 1 ~ 2 rad/sec. What are the values of $M_r$ and $\omega_r$ of the compensated system?

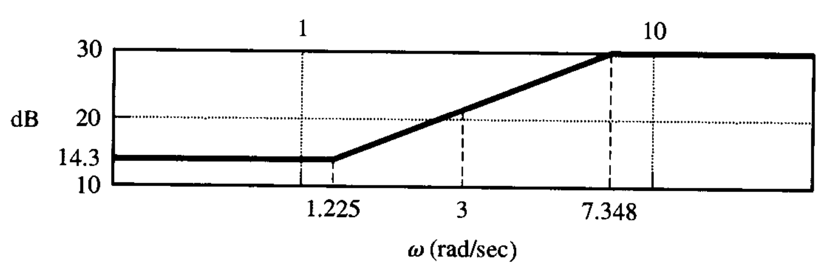



**Figure 9–38** Bode diagram of the lead compensator designed in Problem A-9-5.

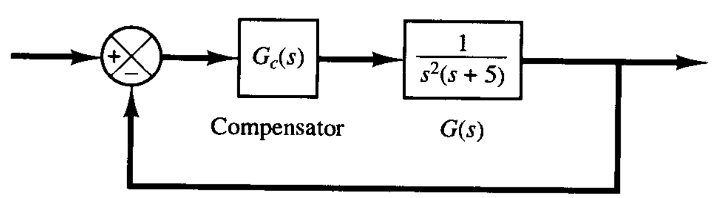



**Figure 9–39** Closed-loop system.

**Solution.** Notice that

$$G_c(j\omega)G(j\omega) = K_c\alpha\left(\frac{Tj\omega + 1}{\alpha Tj\omega + 1}\right)\frac{0.2}{(j\omega)^2(0.2j\omega + 1)}$$

Since the bandwidth of the closed-loop system is close to the gain crossover frequency, we choose the gain crossover frequency to be 1 rad/sec. At $\omega = 1$, the phase angle of $G(j\omega)$ is 191.31°. Hence, the lead network needs to supply 50° + 11.31° = 61.31° at $\omega = 1$. Hence, $\alpha$ can be determined from

$$\sin\phi_m = \sin 61.31° = \frac{1-\alpha}{1+\alpha} = 0.8772$$

as follows:

$$\alpha = 0.06541$$

Noting that the maximum phase lead angle $\phi_m$ occurs at the geometric mean of the two corner frequencies, we have

$$\omega_m = \sqrt{\frac{1}{T}\frac{1}{\alpha T}} = \frac{1}{\sqrt{\alpha}T} = \frac{1}{\sqrt{0.06541}T} = \frac{3.910}{T} = 1$$

Thus,

$$\frac{1}{T} = \frac{1}{3.910} = 0.2558$$

and

$$\frac{1}{\alpha T} = \frac{0.2558}{0.06541} = 3.910$$

Hence,

$$G_c(j\omega)G(j\omega) = 0.06541K_c\frac{3.910j\omega + 1}{0.2558j\omega + 1}\frac{0.2}{(j\omega)^2(0.2j\omega + 1)}$$

or

$$\frac{G_c(j\omega)G(j\omega)}{0.06541K_c} = \frac{3.910j\omega + 1}{0.2558j\omega + 1}\frac{0.2}{(j\omega)^2(0.2j\omega + 1)}$$

A Bode diagram for $G_c(j\omega)G(j\omega)/(0.06541\ K_c)$ is shown in Figure 9–40. By simple calculations (or from the Bode diagram), we find that the magnitude curve must be raised by 2.306 dB so that the magnitude equals 0 dB at $\omega = 1$ rad/sec. Hence, we set

$$20\log 0.06541\ K_c = 2.306$$

or

$$0.06541\ K_c = 1.3041$$

which yields

$$K_c = 19.94$$

The magnitude and phase curves of the compensated system show that the system has the phase margin of 50° and gain margin of 16 dB. Hence, the design specifications are satisfied.

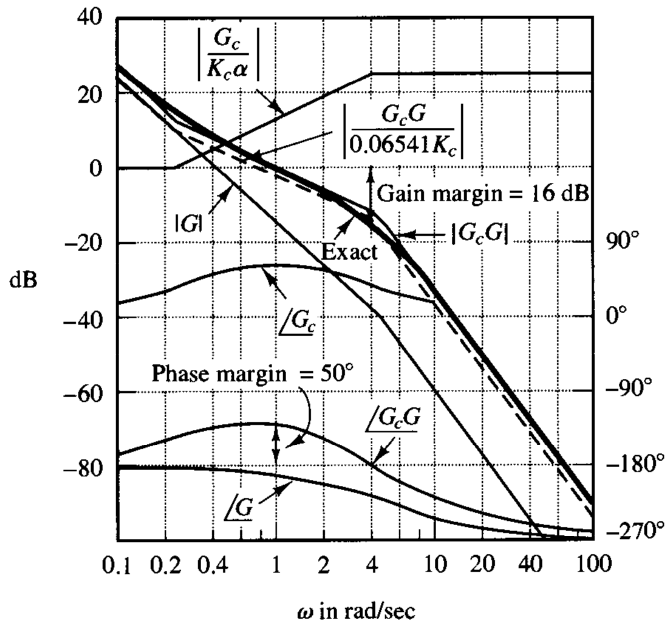



**Figure 9–40**
Bode diagram of the system shown in Figure 9–39.

Figure 9–41 shows the $G_c(j\omega)G(j\omega)$ locus superimposed on the Nichols chart. From this diagram, we find the bandwidth to be approximately 1.9 rad/sec. The values of $M_r$ and $\omega_r$ are read from this diagram as follows:

$$M_r = 2.13 \text{ dB}, \qquad \omega_r = 0.58 \text{ rad/sec}$$

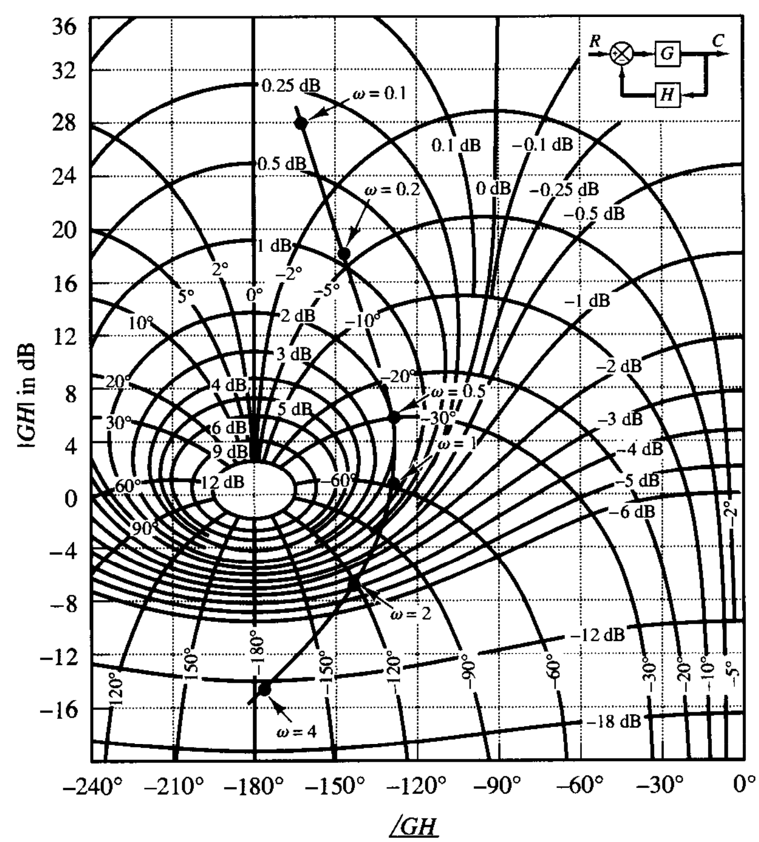



**Figure 9–41**
$G_c(j\omega)G(j\omega)$ locus superimposed on Nichols chart (Problem A-9-6).

**A-9-7.** Referring to Example 9–1, draw Nyquist plots of $G(j\omega)$, $G_1(j\omega)$, and $G_c(j\omega)G(j\omega)$ with MATLAB. (Compare the Nyquist plots obtained here with Figure 9–10.) Write a possible MATLAB program for drawing the Nyquist plots in one diagram. Note that $G(j\omega)$, $G_1(j\omega)$, and $G_c(j\omega)G(j\omega)$ are given by

$$G(j\omega) = \frac{4}{j\omega(j\omega + 2)}$$

$$G_1(j\omega) = \frac{40}{j\omega(j\omega + 2)}$$

$$G_c(j\omega)G(j\omega) = 41.7\frac{j\omega + 4.41}{j\omega + 18.4}\frac{4}{j\omega(j\omega + 2)}$$

**Solution.** A possible MATLAB program for this problem is given in MATLAB Program 9–5. The resulting Nyquist plots are shown in Figure 9–42.

MATLAB Program 9–5

```
%*****Nyquist plots in polar coordinates*****

num1 = [0  0  4];
den1 = [1  2  0];
num2 = [0  0  40];
den2 = [1  2  0];
num3 = [0  0  166.8  735.588];
den3 = [1  20.4  36.8  0];
w = 0.2:0.1:10;
ww = 1.5:0.1:10;
[re1,im1,w] = nyquist(num1,den1,w);
z1 = re1 + i*im1;
r1 = abs(z1);
theta1 = angle(z1);
polar(theta1,r1,'o')
hold on
[re2,im2,ww] = nyquist(num2,den2,ww);
z2 = re2 + i*im2;
r2 = abs(z2);
theta2 = angle(z2);
polar(theta2,r2,'o')
[re3,im3,ww] = nyquist(num3,den3,ww);
z3 = re3 + i*im3;
r3 = abs(z3);
theta3 = angle(z3);
polar(theta3,r3,'x')
title('Nyquist Plots of G(jw), G1(jw), and Gc(jw)G(jw)')
text(0.7,-8.8,'G(jw)')
text(-11.7,-8.7,'G1(jw)')
text(-6.6,-12.3,'Gc(jw)G(jw)')
```

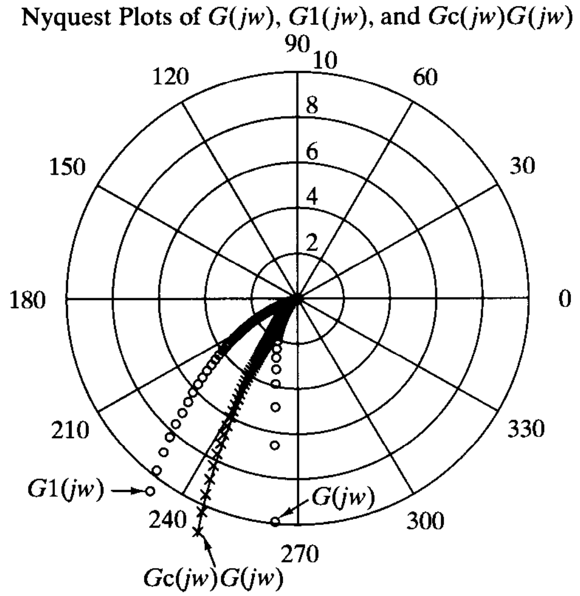



**Figure 9–42**
Nyquist plots of $G(j\omega)$, $G_1(j\omega)$, and $G_c(j\omega)G(j\omega)$.

**A-9-8.** Consider the system shown in Figure 9–43(a). Design a compensator such that the closed-loop system will satisfy the following requirements:

$$\text{Static velocity error constant} = 20 \text{ sec}^{-1}$$

$$\text{Phase margin} = 50°$$

$$\text{Gain margin} \geqq 10 \text{ dB}$$

**Solution.** To satisfy the requirements, we shall try a lead compensator $G_c(s)$ of the form

$$G_c(s) = K_c \alpha \frac{Ts + 1}{\alpha Ts + 1}$$

$$= K_c \frac{s + \dfrac{1}{T}}{s + \dfrac{1}{\alpha T}}$$

(If the lead compensator does not work, then we shall employ a compensator of different form.) The compensated system is shown in Figure 9–43(b).